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R. Pramanik ; R. W. C. P. Verstappen ; P. R. Onck  



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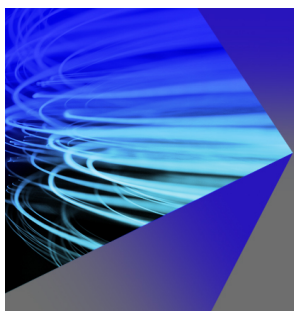
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Cite as: Phys. Fluids **36**, 101302 (2024); doi: [10.1063/5.0226743](https://doi.org/10.1063/5.0226743)

Submitted: 3 July 2024 · Accepted: 1 September 2024 ·

Published Online: 4 October 2024



View Online



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Note: This paper is part of the special topic, Fluids-Structure Interaction.

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ABSTRACT

The omnipresence of fluid–structure interaction (FSI) in biological systems is indisputable—from the vibration of leaves to the locomotion of fish, to the flying of birds, and to the cardiovascular biomechanics; FSI is indeed ubiquitous. Even in stimuli-responsive soft robots that typically operate inside a fluid medium, these physical interactions are prevalent. Therefore, it becomes mandatory to have a thorough understanding of their fully coupled physics involving strong two-way interaction between the solid and fluid domains. Although state-of-the-art computational frameworks and robust numerical techniques have been developed to study their complex physical mechanisms and associated nonlinearities involving multiple spatiotemporal scales, we believe that a timely review of the current development, emerging techniques, and future challenges in computational FSI would further stimulate research along this direction. Therefore, we explore the broad landscape of the myriad research avenues that herald FSI emphasizing their manifold occurrences in biology and advanced soft robotic technologies, while underlining the plethora of numerical techniques adopted to study these fundamental phenomena.

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I. INTRODUCTION

The coupled interaction between a compliant structure immersed within a moving fluid is ubiquitous in a broad range of physical processes in civil engineering,^{1–3} biomechanics,^{4–6} geophysics,^{7–9} aerospace sectors,^{10–14} microfluidics,^{15–20} and soft robotics.^{21–26} In these scenarios, not only do the solid and fluid domains co-exist, but they also affect the motion of each other through a strong two-way interface coupling.^{27–30} These physical interactions become further pronounced when the elastica (i.e., solid domain) undergoes large structural deformations to displace the surrounding fluid (domain).³¹ In turn, the flow field *simultaneously* affects the structural response of the elastica, resulting in fully coupled fluid–structure interaction (FSI).³² When the surrounding medium is a liquid (whose density is orders of magnitude higher compared to air), the FSI is called fluid-elasticity;^{33–35} whereas, if the surrounding fluid is air (or any gaseous mixture that is very sparse and has a low density), its interaction with the elastica is known as aero-elasticity.³⁶

FSI (featuring a soft elastica moving within a dynamic fluid) is omnipresent not only in nature but also in cutting-edge soft robotic technologies (as schematically depicted in Fig. 1).^{37–40} From the

vibration of plants to the fluttering of leaves and swaying of trees,^{41–43} to the cardiovascular and respiratory systems,^{44–46} to the swimming of spermatozoa,^{47–50} to the formation of organoids and self-organizing behavior in functional units of life,^{51–53} to locomotion of aquatic organisms,^{54–56} and to emergent dynamics and collective behavior of bacterial swarms, school of fish, and a flock of birds,^{57–59} FSI is indeed ubiquitous (see Fig. 2).⁶⁰

Even in rigid structures^{83–85} and multi-functional stimuli-responsive soft materials operating inside a fluid,^{86–88} these underlying physical interactions are prevalent (see Fig. 3); e.g., in soft robots actuated using external force fields (such as electrical,^{82,89–91} electronic,^{92–94} hydraulic,^{95–98} pneumatic,^{79,99–102} optical,^{103–106} or magnetic actuation^{107–109}), FSI becomes important.^{110–112} Since the materials used are compliant (for them to maneuver the surrounding fluid and withstand large structural deformations in order to ensure high degrees of freedom), this results in the generation of strong FSI and coupled physics (due to cross-domain energy transduction).¹¹³

Over the years, robust computational models and efficient numerical frameworks have been developed to study these fully

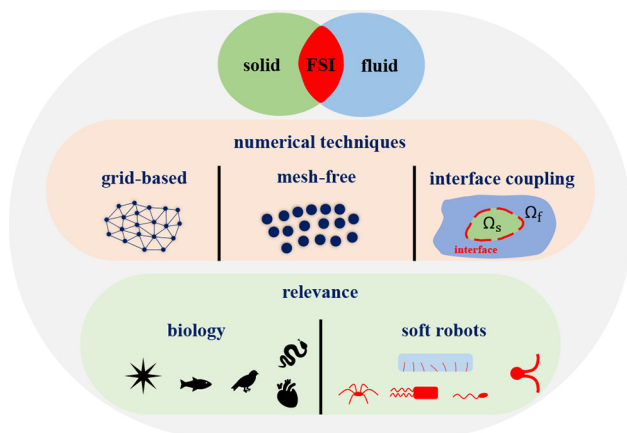


FIG. 1. Summary of this review showcasing various numerical techniques proposed for computational FSI and its extensive occurrences in biology and soft robots.

coupled systems.^{114–125} However, it is extremely challenging when performing high-fidelity three-dimensional FSI simulations. Generally, FSI is modeled using partial differential equations for which closed-form (analytical) solutions are seldom obtainable (owing to their fully coupled and nonlinear nature, inherent complexity, and irregular boundary conditions that are often difficult to implement).^{126–130} Different numerical techniques have, therefore, been proposed to study FSI.^{131–136} For instance, computational solid dynamics (CSD) and computational fluid dynamics (CFD) models have been developed apart from fluid–structure interaction (FSI) models to estimate the performance of turbine blades and bionic flapping wings subjected to

wind loads,^{137–139} or even to predict the stress states of cardiac wall muscles and cerebral aneurysms.^{140–145}

Generally, advanced modeling techniques (such as the updated Lagrangian formulations) are employed to compute the evolution of the solid mesh coordinates when the elastica undergoes large structural deformations.¹⁴⁶ Furthermore, estimating the flow field dynamics requires numerical solutions of the Navier–Stokes (NS) equation involving all the inertial, convective, pressure, and dissipative terms.¹⁴⁷ Thanks to the recent developments and a steep progress in computational power and clusters, numerical simulations have become the preferred choice to study complex problems in comparison to laboratory-based traditional experimental testing.

Although a handful of praiseworthy reviews on a wide range of topics related to FSI have been reported previously focusing on very specific applications in biology and soft robots (see Table I), it is important to note that there is a scarcity of a holistic overview (involving a thorough revisit) of (the milestones achieved in context to) FSI. Thus, we attempt to fill this gap in an elaborative manner by carefully revisiting some of the pioneering efforts and early developments, chronological developments, emerging trends, recent progress, as well as future prospects in this exciting field of interdisciplinary research.

Therefore, we review the broad landscape of myriad research avenues of FSI with special emphasis on their presence in biology and soft robots and their modeling approaches. Not only do we mention the boundless occurrences of FSI in biological systems but also highlight the plethora of numerical techniques adopted to study these physical interactions. We discuss specific case studies of FSI in heart valves and bionic flaps, and also systematically classify the numerical techniques, while underscoring their prominence in nature, active as well as passive elastica. Finally, we mention a few possible areas for further exploration along this field as perspectives.

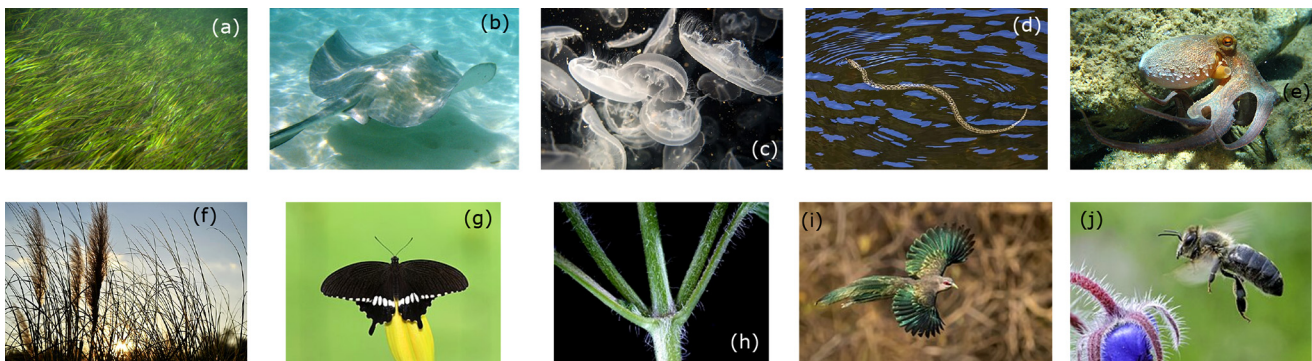


FIG. 2. Examples of FSI in biology. (a) Sea grass between the shore and the reef at Wakaya, Fiji. CC-BY-2.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-2.0 license. It is attributed to Paul Asman and Jill Lenoble. (b) Stingray. CC-BY-2.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-2.0 license. It is attributed to Sam Pullara. (c) Jellyfish. CC-BY-SA-2.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-2.0 license. It is attributed to Eric Kilby. (d) Viperine water snake. CC-BY-SA-4.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Charles J. Sharp. (e) Octopus. CC-BY-SA-3.0-migrated. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-3.0-migrated license. It is attributed to Beckmannjan. (f) A patch of reeds. CC0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC0 license. It is attributed to Ashley St. John. (g) Open wing of *Papilio polytes*. CC-BY-SA-4.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Atanu Bose Photography. (h) Cilia in *Galinsoga quadriradiata*. CC-BY-SA-4.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Gilles Ayotte. (i) Green-billed malkoha. CC-BY-SA-4.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Touhid Biplob. (j) Flying bee. CC-BY-SA-4.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Roy Egloff.

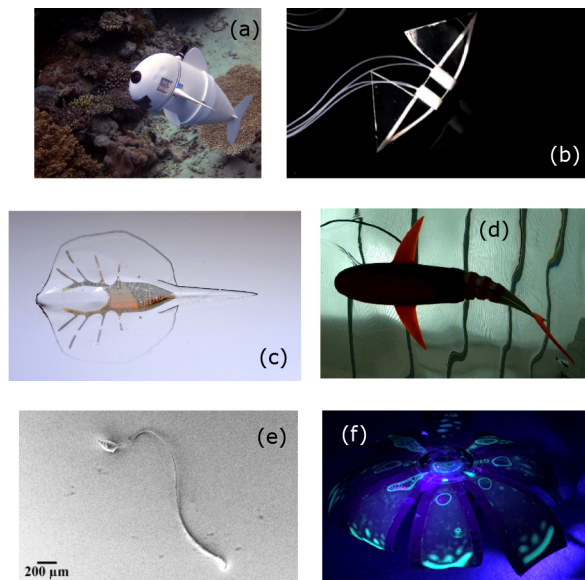


FIG. 3. Examples of FSI in soft robots. (a) Acoustically controlled soft robotic fish.⁷⁸ [Reproduced with permission from Katzschmann *et al.*, *Sci. Robot.* **3**, 16 (2018). Copyright 2018 American Association for the Advancement of Science]. (b) Butterfly-stroke-like soft swimmer.⁷⁹ [Reproduced with permission from Chi *et al.*, *Sci. Adv.* **8**, 46, eadd3788 (2022). Copyright 2022 American Association for the Advancement of Science]. (c) Optically powered tissue-engineered stingray-inspired soft robotic swimmer.⁸⁰ [Reproduced with permission from Park *et al.*, *Science* **353**, 6295, 158–162 (2016). Copyright 2016 American Association for the Advancement of Science]. (d) Electrically driven soft robotic fish.²⁶ [Reproduced with permission from Berg *et al.*, *HardwareX* **12**, e00320 (2022). Copyright 2022 Elsevier]. (e) Magnetically actuated sperm-inspired soft robot.⁸¹ [Reproduced with permission from Khalil *et al.*, *PLoS One* **13**, 11, e0206456 (2018). Copyright 2018 Public Library of Science San Francisco]. (f) Electrically powered jellyfish-inspired soft robot (jellyfishbot).⁸² [Reproduced with permission from Christianson *et al.*, *Front. Robot. AI* **6**, 126 (2019). Copyright 2019 Elsevier].

II. NUMERICAL TECHNIQUES

A. Classification

A systematic classification of the different numerical routes undertaken to study FSI based on the choice of (mesh) grids, mesh-free methods, and interface (domain) coupling is shown in Fig. 4. On the basis of discretization methodology of the fluid domain, fixed¹⁴⁸ and moving grid methods¹⁴⁹ have been developed. Furthermore, based on the methodology of information transmission (about the presence of individual domain and variables of interest), several forcing terms have been formulated, resulting in diffused interface methods [e.g., immersed boundary method¹⁶⁷ and fictitious domain method (FDM)¹⁵⁰] and sharp interface methods (e.g., immersed interface method,¹⁵¹ hybrid Cartesian-immersed boundary method,¹⁵² and immersed finite element method¹⁵³). Furthermore, based on the strength of solid–fluid coupling (and their influence upon each other), there is one-way and two-way FSI. When the structural displacement has negligible influence on the fluid flow, it is considered a one-way FSI; whereas a two-way (i.e., fully coupled) FSI induces a structural deformation large enough to influence the fluid flow, which in turn significantly changes the flow-induced structural deformation (Fig. 5).

A comparison of vortex fields obtained using immersed boundary and immersed interface methods to numerically simulate the viscous flow over a circular cylinder at a Reynolds number of 200 has been reported.¹⁶⁷ Notably, these methods use non-conformal meshes (i.e., the solid and fluid meshes do not have common nodes at the interface). The moving grid methods such as the Arbitrary Lagrangian Eulerian (ALE)¹⁵⁵ and cut-cell methods¹⁵⁶ have conformal body-fitted mesh for transmission of relevant information between solid and fluid domains at the interface through appropriate forcing terms.¹⁵⁷ Although these methods are cumbersome due to re-meshing and smoothing requirements, they are more accurate compared to fixed grid methods.¹⁵⁸ A combination of these methods, known as combined fixed-moving grid-based FSI techniques (e.g., fictitious domain adaptive mesh method,¹⁵⁹ hybrid immersogeometric-ALE FSI method,¹⁶⁰ and operator split Lagrangian–Eulerian method¹⁶¹), has as well been proposed to utilize the individual benefits of fixed and moving grids. However, these approaches are not trivial to implement and also demand higher computational power.

Apart from the grid-based methods, mesh-free methods have also been proposed to study FSI.¹⁶² Particle-based methods such as smoothed particle hydrodynamics¹⁶³ and lattice Boltzmann methods¹⁶⁴ have been developed to study these coupled physical interactions. Furthermore, on the basis of domain coupling at the (fluid–structure) interface, monolithic^{165–167} and partitioned approaches^{168–170} have been adopted. In this regard, the partitioned techniques were either explicitly¹⁷¹ or implicitly coupled.^{172–174}

1. Grid-based methods

Discrete grid-based methods such as finite element method (FEM), finite difference method (FDM), and finite volume method (FVM) have been successful in modeling FSI.¹⁷⁵ For example, FEM uses a mesh with elements of different shapes, such as triangles (or quadrilaterals) in two-dimensional (2D), and tetrahedra (or hexahedra) in 3D. The solution is approximated using basis functions defined over these elements. In areas that have large gradients or feature complex boundary conditions, the mesh is locally refined to capture fine details. It is important to note that the grid is generally unstructured to render flexibility in representing complex geometries and varying element sizes.¹⁷⁶

On the other hand, FDM uses a structured grid, where the points are arranged in a regular pattern (that is often rectangular or cuboidal). Function values are computed at each grid point. Derivatives in the governing partial differential equations (PDEs) are approximated using the function values at these grid points and their neighbors. Although the regularity of the grid makes FDM quite straightforward for implementation, it becomes less flexible in handling complex geometries (compared to FEM and FVM).¹⁷⁷

Finally, FVM uses either structured or unstructured grids, and the computational domain is divided into control volumes (called cells) that are of various shapes (similar to the elements in FEM). This numerical technique focuses on the fluxes across the boundaries of each control volume, wherein the quantities (such as mass, momentum, and energy) are conserved. This method is especially suitable for problems that involve conservation laws (because it inherently enforces conservation at the discrete level).¹⁷⁸ While all three methods are discrete grid-based numerical techniques, they differ in their grid structure, implementation, and applicability.

TABLE I. A chronologically listed summary of previous reviews focusing on FSI with applications in biology and soft robots.

Year	Reference	Primary focus
2001	Dowell <i>et al.</i> ⁶⁰	Fundamental understanding of FSI and stability analysis in engineering and biomedical applications
2007	Hron <i>et al.</i> ⁴	FSI in biomechanics using fully coupled Arbitrary Lagrangian-Eulerian formulation
2016	O'Connor <i>et al.</i> ⁶¹	Usage of lattice Boltzmann method in micro-circulation and microfluidic applications
2016	Parameshwaran <i>et al.</i> ⁶²	FSI and flow-induced vibrations of moving/dynamic structures subjected to wind flows
2016	Wang <i>et al.</i> ⁶³	FSI in flapping-powered propulsion: biomimetic design of artificial flyers and swimmers
2016	Huang <i>et al.</i> ⁶⁴	State-of-the-art developments in FSI modeling: biological and biomechanical applications
2017	Boyer <i>et al.</i> ⁶⁵	FSI in biomimetic robots with soft structures: applications during flying and swimming
2017	Liu <i>et al.</i> ⁶⁶	Unsteady bio-fluid dynamics: leading-edge vortices and airflow mechanisms in flexible wings/foils
2019	Faizal <i>et al.</i> ⁴⁵	FSI simulations for patients with sleep-related breathing disorders with obstructive sleep apnea
2020	Griffith <i>et al.</i> ⁶⁷	Immersed methods for large deformation FSI with applications at high Reynolds numbers
2021	Mazhar <i>et al.</i> ⁶⁸	Mesh-free particle-based methods for FSI problems and their classifications
2021	Toma <i>et al.</i> ⁶⁹	FSI modeling of dynamical biological systems using smoothed-particle hydrodynamics
2022	Wang <i>et al.</i> ⁷⁰	FSI of bio-inspired flexible slender structures: deformable flapping foils, membranes, and filaments
2022	Abbas <i>et al.</i> ⁷¹	Assessment of different FSI methods for cardiac mechanics, aortic and mitral heart valve simulations
2022	Zhang <i>et al.</i> ⁷²	FSI models, vortex dynamics, and key challenges during flying and swimming
2022	Le <i>et al.</i> ⁷³	FSI simulations of cardiac mechanics: valvular structures and human left heart anatomy
2022	Ashraf <i>et al.</i> ⁷⁴	FSI in obstructive sleep apnea: pharyngeal airway deformation mechanisms and airflow dynamics
2022	Mourato <i>et al.</i> ⁷⁵	Computational modeling and simulations of blood-vessel FSI in aortic aneurysms and dissections
2023	Syed <i>et al.</i> ⁷⁶	Cardiovascular modeling: FSI between blood, arterial walls, heart walls, and valves
2023	Jung <i>et al.</i> ²⁹	Biological and bio-inspired FSI problems with new insights into natural systems
2023	Kuchumov <i>et al.</i> ⁷⁷	FSI simulations of aortic valve surgery: valve reconstruction methods and surgical outcomes

The entire computational domain is discretized with cells (or elements) in grid-based methods (see Fig. 6). Numerical simulations of large deformation FSI face a severe challenge while generating the appropriate stiffness model required to predict excessive mesh deformation. Generally, body-fitted (conformal) meshes undergo a degradation of the mesh quality giving rise to spurious numerical oscillations.¹⁷⁹ In the ALE method, the Eulerian fluid mesh moves following the Lagrangian structural mesh. For small deformations and minimal structural displacements, stable and convergent numerical results are obtainable. However, when the structural displacements are large, moving-grid methods face serious obstacles and numerical challenges.

Another body-fitted mesh method in the form of a *space-time* finite element method has been developed for similar problems involving moving boundaries and interfaces.¹⁸⁰ Hybrid meshes have been used for this geometrically conserving ALE method for the strongly coupled FSI. Mesh-smoothing algorithms have been implemented utilizing techniques such as swapping of faces or edges, changing the number of vertex, and changing mesh connectivity.¹⁸¹ Linear and non-linear mesh smoothing algorithms were used for mesh optimization, re-meshing, and adaptive meshing. For structures undergoing large displacements, the extended ALE method has been proposed,¹⁸² where a variational mesh optimization algorithm was used whenever there was a mesh degradation (or mesh degeneration) due to large structural

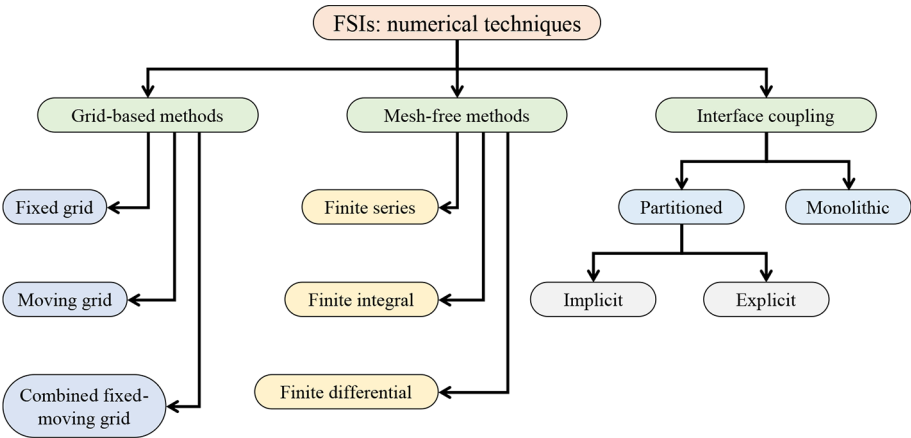


FIG. 4. A schematic representation of different numerical techniques and classification of FSI schemes based on grid-based methods, mesh-free methods, and fluid-structure interface coupling.

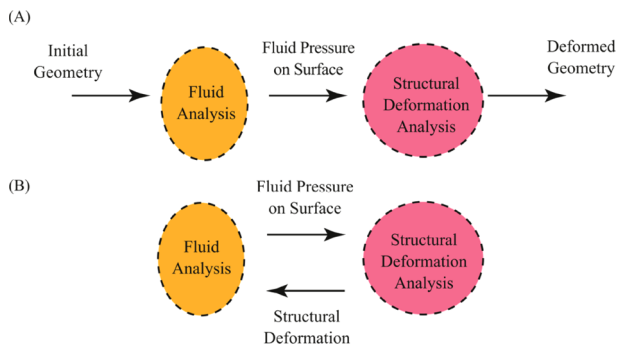


FIG. 5. Schematic representation of (a) one-way and (b) two-way FSI.¹⁵⁴ [Reproduced with permission from Musharaf *et al.*, *Micromachines* **15**(7), 897 (2024). Copyright 2024 MDPI].

deformations. The mesh alignment with the structure interface was achieved via a constraint, resulting in unconditional stability, convergence, and accuracy.¹⁸²

Three-dimensional simulations involving an immersed boundary flow solver for the incompressible NS equation and nonlinear finite element solid mechanics solver for hyperelastic Saint Venant–Kirchhoff model have been performed to understand the aerodynamics of elastic insect wings.¹⁸³ In this study, a strong coupling was incorporated using a modular partitioned scheme, along with a Chorin projection and Crank–Nicolson scheme. The inherent disadvantage with these methods is that an algorithm is required to move (or re-define) the mesh in the Eulerian fluid along with the structural boundary points. To address this, non-conforming meshes have been used to tackle the interface conditions using forcing terms in the solid and fluid governing equations.¹⁸⁴

Based upon the approach of transmission of the solid domain presence to the fluid domain and vice versa, fixed grid methods are distinguished. In the fixed-grid methods, the structural domain is represented using a Lagrangian description (data points), and the fluid domain has a fixed-background Eulerian mesh description. Two such

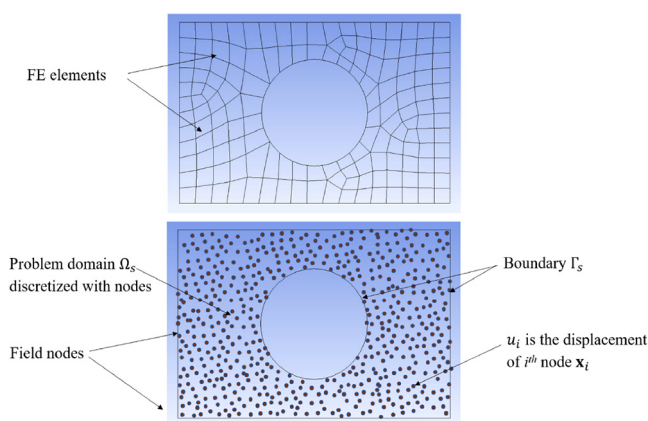


FIG. 6. Discretization of the computational domain using a grid-based methods (top) and a mesh-free method (bottom).⁶⁸ [Reproduced with permission from Mazhar *et al.*, *Eng. Anal. Boundary Elements* **124**, 14–40 (2021). Copyright 2021 Elsevier Ltd.].

promising methods are IBM¹⁸⁵ and the fictitious domain method (FDM).¹⁸⁶ They use simple Cartesian structured grids even for complex geometries, and these are particularly effective while dealing with moving boundaries. The forcing term takes into account the presence of soft elastica inside the fluid domain. While for the IBM, this imposes the Dirichlet boundary condition at the immersed body surface, in the case of FDM, the forcing term imposes a rigidity condition inside the immersed body.

While the continuous forcing method was developed by Peskin,¹⁸⁷ the discrete direct forcing technique based upon local flow reconstruction¹⁸⁸ and projection approach¹⁸⁹ was developed subsequently by other researchers. These methods have the advantage of *zero mesh distortion* since these are non-body-conformal meshes. Therefore, complex geometries can be handled even when the structure undergoes large deformations. While this may be true, implementing the boundary conditions at the fluid–solid interface is not trivial. In this regard, a hybrid approach that combines the advantages of both IBM and FDM has been developed (that showed an excellent agreement with experiments) for a free-falling particle.¹⁹⁰

2. Mesh-free methods

Often, mesh-free methods are adopted to solve FSI problems instead of classical grid-based methods (e.g., finite element method).¹⁹¹ The fundamental difference lies in the connection between each node in the computational domain. Mesh-free methods do not require a mesh (or a defined grid) to establish a connection between the Lagrangian points. Rather, they depend on the interaction of each node with all its neighbors.^{192–194} Although this results in an increased computational cost, the flexibility, adaptability, and extensibility of these methods set them apart for applications such as large deformation FSI or fracture mechanics (where the connectivity of the computational mesh is often difficult to maintain, or the mesh distortion exceeds reasonable limits).^{195–197} From a computational perspective, coupling of particle-based structure and fluid models results in an entirely Lagrangian mesh-free particle-based solver. The solution steps in the mesh-free formulation have been shown in Fig. 7.

One of the oldest mesh-free methods, smoothed particle hydrodynamics (SPH),^{198,199} solves these problems by considering Lagrangian (data) points as physical particles (with mass and density) that move throughout the computational domain over time and, therefore, evolve.²⁰⁰ These point masses carry some physical properties with them (such as spatial coordinates and velocity) that have smoothed-out properties defined using appropriate kernels (see Fig. 8), which are equivalent to Dirac-Delta functions in discrete form.^{201–203} Although the advantages of SPH outnumber their disadvantages, the potential challenges are modeling the boundary conditions and tension instabilities.²⁰⁴

This has subsequently led to the development of other mesh-free methods (for FSI modeling) such as diffuse element method (DEM),²⁰⁵ local Petrov–Galerkin (MLPG) method,^{206,207} element free Galerkin (EFG) method,²⁰⁸ reproducing kernel particle method (RKPM),^{209,210} moving particle semi-implicit (MPS) method,²¹¹ dissipative particle dynamics (DPD),²¹² SPH-lattice-spring model (LSM),²¹³ finite pointset method (FPM),²¹⁴ and material point method (MPM).²¹⁵ The latest developments in mesh-free methods have been the space-time mesh-free collocation method (STMCM)²¹⁶ and the mesh-free interface-finite element method (MIFEM).²¹⁷ In general, the

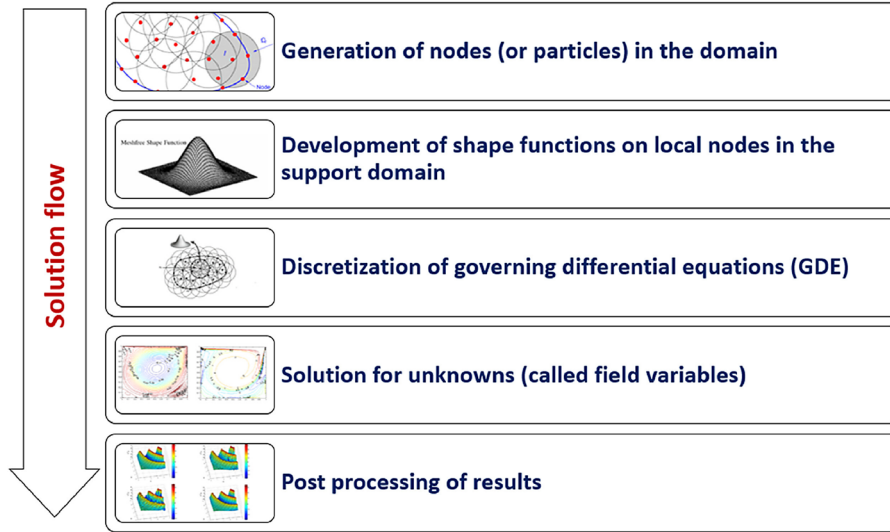


FIG. 7. Solution methodology in mesh-free formulation.⁶⁸ [Reproduced with permission from Mazhar *et al.*, Eng. Anal. Boundary Elements **124**, 14–40 (2021). Copyright 2021 Elsevier Ltd.].

mesh-free methods lack the Kronecker-delta property that makes it difficult to enforce the essential boundary conditions (at least compared to the finite element method, where they are imposed directly).²¹⁸

3. Interface coupling

On the basis of interface coupling between the structural and fluid domains, FSI is classified into monolithic (direct) and partitioned approaches.²¹⁹ The former has a unified approach for formulation and solution of both the fluid and structure equations, which allows a unified space-time discretization. However, this involves a high computational cost and is often prone to numerical errors. In the partitioned approach, the fluid and structure equations are solved iteratively, via either a weak explicit coupling (fluid and structure problem at a time) or a strong implicit coupling (sub-time level iterations). However, issues relating to convergence, stability, and efficiency ought to be accounted for, while using this approach. While the explicit interface coupling technique is mostly unsuitable and often unstable, the implicit coupling is generally stable and computationally cheaper.

4. Mathematical insights

Typically, computational FSI techniques involve methods that couple fluid dynamics and structural mechanics to analyze physical

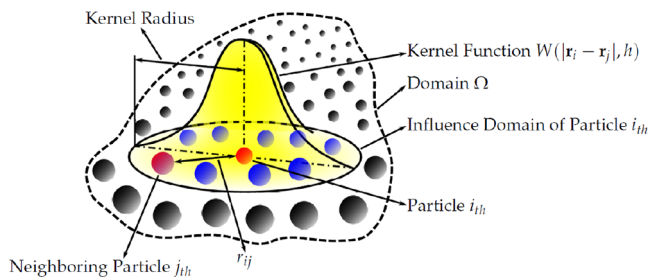


FIG. 8. SPH represented using a kernel approximation.⁶⁹ [Reproduced with permission from Toma *et al.*, Biology **10**, 185 (2021). Copyright 2021 MDPI].

(mechanical) interactions between fluid and solid domains (see Fig. 9). Here is a brief abridgment of the different computational FSI techniques and their associated equations:

- **Monolithic approach:** This can be described as a numerical methodology that solves the fluid and structure equations simultaneously within a single system of equations.²²⁰ Generally, NS equations are used to model the incompressible fluid [(1) and (2)], whereas linear or nonlinear elasticity equations account for the solid [(3) and (4)]. Although the monolithic coupling is often challenging to implement due to complexity, it offers direct coupling between fluid and structure to make the analysis robust:

$$\frac{\partial v_f}{\partial t} + (v_f \cdot \nabla) v_f = -\frac{1}{\rho_f} \nabla p + \nu \nabla^2 v_f, \quad (1)$$

$$\nabla \cdot v_f = 0, \quad (2)$$

$$\sigma = C : \varepsilon, \quad (3)$$

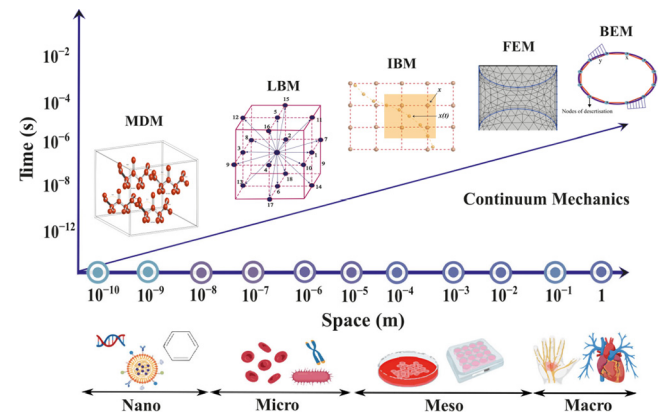


FIG. 9. Length and time scales of different computational FSI techniques.¹⁵⁴ [Reproduced with permission from Musharaf *et al.*, Micromachines **15**(7), 897 (2024). Copyright 2024 MDPI].

$$\rho_s \frac{\partial^2 u_s}{\partial t^2} = \nabla \cdot \sigma + f. \quad (4)$$

Here, σ is the stress tensor, C is the stiffness matrix, ε is the strain tensor, ρ_f is the fluid density, ρ_s is the solid density, v_f is the fluid velocity, u_s is the solid displacement vector, and f is the external force.

- **Partitioned approach:** This methodology uses the same set of equations (1)–(4); however, it solves fluid and structural problems separately and iterates between the two solutions to achieve convergence.²²¹ It is more flexible and easier to implement and is often used with explicit time-stepping methods.
- **Strongly coupled (two-way) approach:** This approach iteratively couples fluid and structural solvers, exchanging information at each time step.²²² Although this allows for full interaction between fluid and structure, this approach is often computationally intensive. The fluid (1) and solid (4) equations are now written, respectively, as

$$\rho_f \frac{\partial v_f}{\partial t} + \rho_f (v_f \cdot \nabla) v_f = -\nabla p + \mu \nabla^2 v_f + f_s, \quad (5)$$

$$\rho_s \frac{\partial^2 u_s}{\partial t^2} = \nabla \cdot \sigma + f_f. \quad (6)$$

Here, f_s and f_f are the fluid force exerted by the structure and the structural force exerted by the fluid, respectively.

- **Weakly coupled (one-way) approach:** This approach solves the fluid and structural problems separately with minimal interaction, often with the structure being fixed and the fluid flow affecting it (rather) passively. This approach is suitable for problems where the structural deformation is small compared to the fluid motion. Therefore, the fluid (5) and solid (6) equations are now written, respectively, as

$$\rho_f \frac{\partial v_f}{\partial t} + \rho_f (v_f \cdot \nabla) v_f = -\nabla p + \mu \nabla^2 v_f, \quad (7)$$

$$\rho_s \frac{\partial^2 u_s}{\partial t^2} = \nabla \cdot \sigma + f_f. \quad (8)$$

- **ALE approach:** This methodology combines the Eulerian and Lagrangian descriptions to handle large deformations and mesh movement.²²³ In the ALE framework, the domain or mesh can move, and this requires modifications to account for the mesh motion. The fluid momentum equation (1) now change to Eqs. (9) and (10), while the continuity equation (2) changes to Eq. (11) as

$$\frac{\partial v_f}{\partial t} + (v_f - v_m) \cdot \nabla v_f = -\frac{1}{\rho_f} \nabla p + \nu \nabla^2 v_f \quad (9)$$

with mesh movement handled via the ALE frame:

$$\frac{\partial x}{\partial t} = v_m. \quad (10)$$

Here, x is the mesh position and v_m is the mesh velocity:

$$\frac{\partial(\rho v_f)}{\partial t} + \nabla \cdot (\rho v_f v_f) = \rho v_m \cdot \nabla v_f - \nabla \cdot T + f. \quad (11)$$

Here, T is the stress tensor, and f represents body forces (e.g., gravity). The displacement equation for the solid domain remains

the same as Eq. (6). Furthermore, the solid–fluid coupling is achieved using either force balance (dynamic coupling) or displacement matching (kinematic coupling), as discussed next.

- **Coupling conditions:** In FSI, the coupling conditions ensure the continuity of stresses and displacements across the fluid–structure interface:²²⁴ kinematic (12) and dynamic continuity (13):

$$v_f = \frac{\partial u_s}{\partial t}, \quad (12)$$

$$\sigma \cdot n = \tau \cdot n. \quad (13)$$

Here, n is the normal vector to the interface and τ is the fluid stress tensor.

- **Immersed boundary method (IBM):** The force exerted by the boundary on the fluid (i.e., f_s) is computed based on the boundary's configuration and its interaction with the fluid.²²⁵ This force is often calculated using an elastic model for the boundary as

$$f_s = K(X - X_0) + D \frac{\partial X}{\partial t}. \quad (14)$$

Here, K is the stiffness matrix and D is the damping coefficient. X and X_0 are the current and reference position of the boundary points, respectively. The force from the boundary to the fluid (i.e., f_f) is spread using a delta function δ_d as

$$f_f(x) = \int f_s(X(s, t)) \delta_d(x - X(s, t)) ds. \quad (15)$$

The velocity of the boundary is interpolated from the fluid velocity field using the same delta function as

$$\frac{dX(s, t)}{dt} = \int v(x, t) \delta_d(x - X(s, t)) dx. \quad (16)$$

Here, δ_d is the discrete delta function, often approximated in numerical simulations. This is a key component in IBM and is used to ensure that the force and velocity are accurately transferred between the Eulerian fluid grid and the Lagrangian boundary points. A common choice for the same is a smoothed, compactly supported function such as

$$\delta_d = \frac{1}{h^d} \phi\left(\frac{x}{h}\right). \quad (17)$$

Here, h is the grid spacing, d is the dimension of the problem (i.e., 2D or 3D), and ϕ is a smoothing kernel function, such as a Gaussian or a polynomial spline. The coupled fluid–structure system is typically solved using a time-stepping scheme, such as an explicit or semi-implicit method. It is important to note that IBM provides a versatile and powerful framework for simulating complex fluid–structure interactions, especially in biological and engineering applications where the structures may be flexible or undergo large motions.

- **Fictitious domain method (FDM):** The fundamental idea behind this methodology is to embed the physical domain (where the structure and fluid interact) within a larger, simpler computational domain (the *fictitious* domain), allowing for the use of regular grids and standard numerical solvers.²²⁶ Here, the fluid and solid domains are coupled using a Lagrange multiplier λ to impose the no-slip condition on the fluid–structure interface. λ is

used to enforce this constraint by adding it to the fluid momentum equation:

$$\frac{\partial v_f}{\partial t} + (v_f \cdot \nabla) v_f = -\frac{1}{\rho_f} \nabla p + \nu \nabla^2 v_f + f_s + G\lambda. \quad (18)$$

Here, G is a suitable operator that spreads the Lagrange multiplier over the fluid–structure interface. The fictitious domain approach involves embedding the actual domain within a larger, regular computational domain. The fluid and structure are solved together in this larger domain, with the structure's influence incorporated using the Lagrange multiplier method. This allows the use of a single grid for both fluid and structure, simplifying the numerical implementation.

- **Lattice Boltzmann method (LBM):** This methodology is well-suited for FSI problems due to its ability to handle complex boundary conditions and moving geometries. The fundamental equation in LBM is the lattice Boltzmann equation,^{227,228} describing the evolution of the density distribution functions f_i at discrete lattice points and times:

$$f_i(x + ce_i \Delta t, t + \Delta t) - f_i(x, t) = -\frac{1}{\tau} (f_i(x, t) - f_i^{eq}(x, t)) + f_s. \quad (19)$$

Here, f_i is the distribution function along the direction e_i at position x and time t , e_i are the discrete velocity directions, Δx is the lattice grid spacing, Δt is the time step, $c = \Delta x / \Delta t$, τ is the dimensionless relaxation time, and f_i^{eq} is the equilibrium distribution function (and depends on the lattice model used). The macroscopic variables such as density ρ and velocity v are obtained by summing over the distribution functions:

$$\rho(x, t) = \sum_i f_i(x, t), \quad (20)$$

$$\rho v(x, t) = \sum_i f_i(x, t) c e_i. \quad (21)$$

In IBM, f_s is incorporated as a source term in the LBM equation, where the force distribution is spread to nearby fluid nodes. LBM for FSI involves solving the lattice Boltzmann equation on a discrete lattice while coupling the fluid and structural dynamics through appropriate boundary conditions and force calculations.

- **Smoothed particle hydrodynamics (SPH):** This is a mesh-free, Lagrangian computational method used for simulating fluid flows, and it is particularly useful for handling FSI due to its flexibility with moving boundaries.²²⁹ In SPH, fluids and structures are represented as a collection of particles, each carrying properties such as mass, position, velocity, and other physical quantities. In SPH, the properties at any point are estimated using the properties of nearby particles weighted by a smoothing (kernel) function $W(r, h)$ defined as

$$W(r, h) = \frac{\chi}{h^d} w\left(\frac{|r|}{h}\right). \quad (22)$$

Here, r is the distance vector between particles, χ is the normalization constant, and w is a dimensionless kernel function. The density ρ_i of a particle i is approximated as

$$\rho_i = \sum_j m_j W(r_{ij}, h). \quad (23)$$

Here, m_j is the mass of particle j , and $r_{ij} = r_i - r_j$ is the position vector from particle j to particle i . The momentum conservation equation for particle i in SPH form is

$$\frac{dv_i}{dt} = - \sum_j m_j \left(\frac{P_i}{\rho_i^2} + \frac{P_j}{\rho_j^2} + \Pi_{ij} \right) \nabla_i W(r_{ij}, h) + f_i^{ext}. \quad (24)$$

Here, v_i is the velocity of particle i . P_i and P_j are pressures of particles i and j , respectively. $\nabla_i W(r_{ij}, h)$ is the gradient of the kernel function with respect to the position of particle i . Π_{ij} is the artificial viscosity term that accounts for shocks and viscous effects. f_i^{ext} is the external force acting on particle i . On the other hand, the motion of the structure is governed by

$$\frac{du_i^{str}}{dt} = v_i^{str}, \quad \frac{dv_i^{str}}{dt} = \frac{1}{m_i^{str}} (f_i^{str} + f_i^{int}). \quad (25)$$

Here, u_i^{str} , v_i^{str} , and m_i^{str} are the displacement, velocity, and mass of the structural particle i , respectively. f_i^{str} is the force due to the fluid on the structure. f_i^{int} is the internal force due to structural stiffness and damping. The interaction force between fluid and structure is determined by

$$f_i^{str} = - \sum_j m_j \left(\frac{P_j}{\rho_j^2} + \Pi_{ij} \right) \nabla_i W(r_{ij}, h). \quad (26)$$

In SPH for FSI, the fluid and structure are both modeled as particles, and their interaction is described by coupling the SPH equations for fluid dynamics with the equations for structural dynamics. This mesh-free approach allows for the simulation of complex, time-varying geometries and interactions.

B. Previous and current trends

1. Biomedical sciences

A pioneering study on the two-way nonlinear FSI coupling of blood flow through compliant vessels was conducted around early 21st century using a multi-scale computation model.²³⁰ Around the same time, numerical solutions of NS equations and coupled FSI were obtained for incompressible flows over a circular cylinder (see Fig. 10).²³¹ Subsequently, a computational model for FSI in aortic valves was developed using Lagrange multiplier-based fictitious domain method.²³² The large leaflet motion and incompressible (and iso-thermal) NS equation were solved by enforcing no-slip constraint and traction continuity at the blood–valve interface (see Fig. 11). Specifically, the ALE method was used to solve free-boundary problems through a reference domain formulation and an artificial domain velocity.²³³ Similarly, a combined fictitious domain approach was undertaken using a Lagrangian description for the nonlinear solid and an Eulerian description for the surrounding fluid.²³⁴ Here, a non-conforming solid–fluid moving-mesh with adaptive meshing algorithm was used to account for the large structural displacements.

Motivated by the heart valve dynamics, a 3D computational model has been developed to study the interaction of an incompressible viscous fluid with an immersed thin elastic structure.²³⁵ In subsequent studies, simulations of heart valves and their large (dynamic) deformations were predicted using IBM.^{236,237} Furthermore, the pulsating nature of blood flow generated viscoelastic waves that

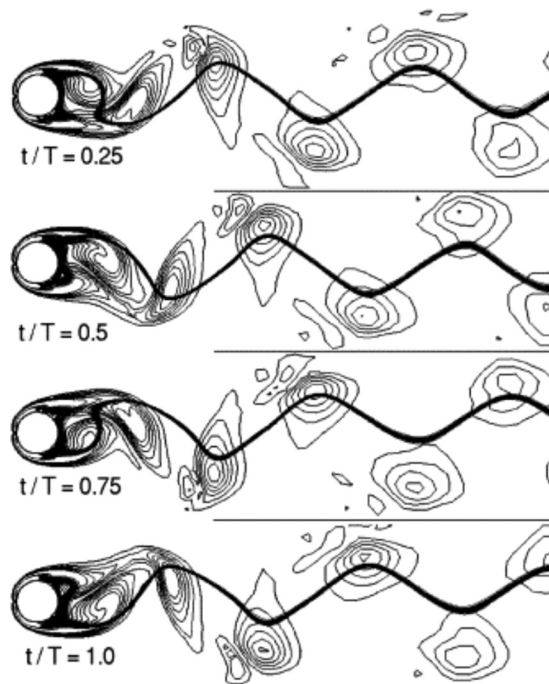


FIG. 10. A pioneering study showing the flow over a circular cylinder at a Reynolds number of 200 at various non-dimensional times: particle traces and vorticity magnitude contours.²³¹ [Reproduced with permission from Kiris *et al.*, *Comput. Fluids* **30**, 7–8, 829–851 (2001). Copyright 2001 Elsevier Ltd.].

propagated through compliant arterial walls; this fully coupled FSI was modeled using Kelvin–Voigt Koiter shell elements (for arterial walls) and incompressible NS equations (for blood).²³⁸ A kinematically coupled time-splitting modular algorithm was used for FSI during blood flows through arteries.²³⁹ Here, the Robin-based semi-implicit coupling, Nitsche’s method, and Chorin–Temam projection schemes were used for partitioned FSI through explicit splitting of viscous and convective terms.

FSIs between pulsatile blood flow through viscoelastic arteries have been studied using NS equations modified with the ALE framework;²⁴⁰ specifically, the Moens–Korteweg equation was used to relate the pulse wave velocity and the arterial wall properties. In another study, FSI was modeled using the fixed grid fictitious domain technique,²⁴¹ wherein the fluid component was modeled using unsteady NS equation with non-Newtonian rheology (for the blood flow) and incompressible hyperelastic Neo-Hookean model (for the arteries). Here, the no-slip condition at the fluid–solid interface and traction continuity were reinforced through the use of Lagrange multipliers, which did not necessitate the requirement of re-meshing (or mesh refinement) at every time increment. Subsequently, the existing FSI framework was extended to incorporate a solid-rigid body contact algorithm to account for appropriate estimation of the contact force.²⁴² The two-dimensional FSI between a deformable solid and the surrounding fluid was studied and represented through streamline plots (see Fig. 12). An extended finite element formulation had been proposed for similar large deformation FSI,²⁴³ where the incompressibility constraint was considered in the case of Saint Venant–Kirchhoff

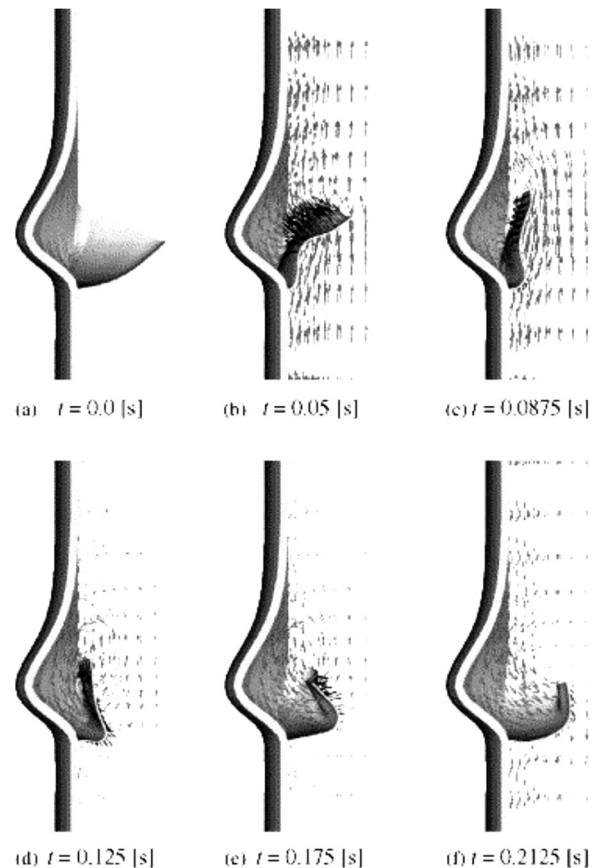


FIG. 11. (a)–(f) Schematic representation of velocity fields depicting FSI between heart valves and blood flow for six successive points in time.²³² [Reproduced with permission from Hart *et al.*, *J. Biomech.* **36**, 1, 103–112 (2003). Copyright 2003 Elsevier Ltd.].

hyperelastic solid. Here, a partitioned scheme was used along with Aitken’s acceleration for a faster convergence.

Along similar lines, FSI in blood flow resulting in non-zero longitudinal structural displacement has been modeled using Lie-splitting fluid–structure sub-problem decoupling approach with increased accuracy and performance.²⁴⁴ A partitioned scheme for fluid–composite structure interaction problems was reported for coupled nonlinear FSI problem with unconditional stability using Robin boundary condition.²⁴⁵ In this study, linear elasticity theories was used for the thick structure, while Koiter shell was used for the thin membrane; modular and operator-splitting schemes were used for spatiotemporal discretization. A higher-order combined DG-ALE partitioned approach has been undertaken to solve fully coupled problems with viscous incompressible fluids and elastic structures.²⁴⁶ Here, a domain decomposition (DD) technique was used for complete spatiotemporal discretization along with Robin–Neumann boundary conditions. The structural component was modeled as thin blade and plate elements, while the fluid was modeled using incompressible unsteady full NS equations.^{247,248} A re-meshing scheme was adopted where the fluid mesh followed the solid mesh at the interface for every time step. The

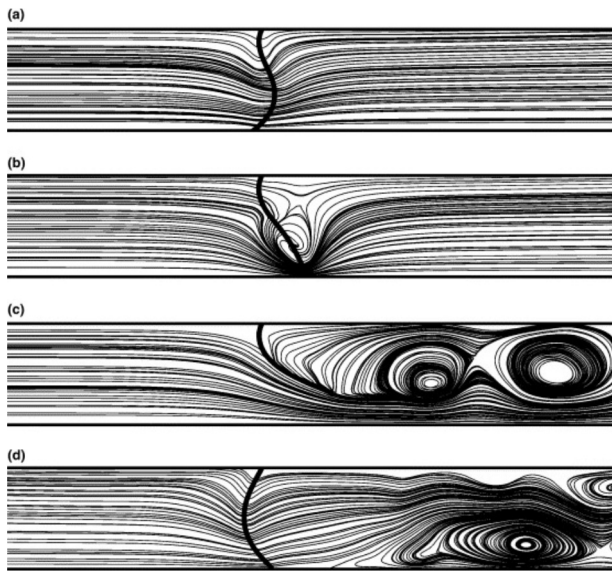


FIG. 12. Numerical simulations of two-dimensional FSI shows the streamline plots and position of the solid slab at different time instances:²⁴² (a) $t/T = 0.1$, (b) $t/T = 0.12$, (c) $t/T = 0.3$, and (d) $t/T = 0.59$. [Reproduced with permission from van Loon *et al.*, *J. Comput. Phys.* **217**, 2, 806–823 (2006). Copyright 2006 Elsevier].

method shows high efficiency, robustness, and convergence with unconditional stability.^{249,250}

In Fig. 13, a schematic representation of the cardiovascular system has been depicted that shows the heart valves and other components in two different planes. Pulsatile flows exhibit complex FSI in a bi-leaflet mechanical heart valve, and this has been studied using a strongly coupled partitioned approach (with under-relaxation) in conjunction with the Aitken acceleration technique.^{251,252} With an objective to model the dynamics of bio-prosthetic heart valves, a non-linear large deformation finite element method (for soft tissues) has been coupled with a sharp-interface immersed boundary incompressible Navier–Stokes solver (for fluids).²⁵³ Here, a strongly coupled partitioned approach was adopted that was stabilized with under-relaxation and the Aitken acceleration technique. As an endeavor to mimic biological leaflet valve, two deformable filaments have been installed inside a channel flow and the resulting FSI was analyzed using the immersed boundary method.²⁵⁴ An unified continuum ALE finite element method with moving mesh monolithic approach has been developed for modeling 3D FSI of heart valves.²⁵⁵ FSI in abdominal aortic aneurysm was studied to understand the influence of asymmetry and wall thickness upon its biomechanical environment (flow and wall dynamics) using fully coupled models;²⁵⁶ here, linear hexahedral elements used for meshing were imported into a commercial finite element code for transient flow condition analysis.

Numerical simulations of 2D stented aortic valve at high Reynolds number flows have been reported using Taylor–Hood (TH) finite elements instead of Crouzeix–Raviart (CR) family.²⁵⁷ In TH family, velocity and deformation are quadratic, while pressure is linear. However in CR family, velocity, deformation and pressure are all quadratic, which increases the computational cost. FSI simulations of bio-prosthetic heart valves have been carried out using Fung-type material

models.²⁵⁸ Kirchhoff–Love hyperelastic thin shell elements have been used for the structure, while the fluid (blood) has been modeled using NS equations and ALE formulations. In this work, an updated Lagrangian framework has been coupled with an Eulerian fluid one to result in a monolithic FSI with fixed grid method. Recently, FSI in tricuspid and bicuspid aortic valves have been studied using coupled lattice Boltzmann method (LBM)-FE techniques.²⁵⁹

2. FSI in engineering

Airflow over wings can induce flutter, analyzed by computational FSI for aeroelastic behavior and wing deformation.^{260,261} NACA airfoil simulations with CFD and Reynolds averaged Navier–Stokes (RANS) models matched experimental data.²⁶² CFD-FEM methods were compared for mesh morphing and structural mode integration.²⁶³ Flutter analysis used RANS-based CFD and a structural code,²⁶⁴ while mesh morphing methods improved computation efficiency.²⁶⁵ ALE framework simulations for a double wedge airfoil employed fractional-step methods.²⁶⁶ Studies also explored wind turbine blade vibrations, aerodynamic-driven topology optimization, and FSI models for rocket bodies.^{267–272} The dynamics of a flexible flapping wing were analyzed using a nonlinear FSI model, revealing bifurcations to quasi-periodic motion.^{273,274} Additional studies investigated vortex flow fields, flexible filament structures, and energy harvesting from vortex-induced vibrations.^{275–278} The impact of a flexible aft-tail on wake dynamics showed improved flow regularization.²⁷⁹

FSI between ship hulls and water waves impacts stability and performance through hydrodynamic forces.²⁸⁰ A partitioned computational model was developed to include hydrodynamic recovery forces.²⁸¹ Flexible marine risers experiencing vortex-induced vibrations were analyzed using a Multi-Iterative Coupling procedure for improved stability and reliability in FSI simulations.^{282,283} Hydroelastic characteristics of flexible marine propellers were studied with BEM-FEM and CFD-FEM models.^{284,285} Lagrangian mesh-free solvers and multi-resolution particle methods were used to study hydro-elastic slamming and fatigue in marine structures.^{286–288} Flexible hydrofoils were examined for vibration characteristics under static and transient loading conditions.²⁸⁹ A study on solitary wave impact on a horizontal plate used moving particle semi-implicit and FEM for force and pressure distribution analysis.²⁹⁰ Additionally, FSI modeling of flat plate water entry and marine rudder dynamics in the wake of a propeller were investigated.^{291,292}

Wind or water flow can cause oscillations or resonance in bridges, impacting structural integrity. Computational FSI models predict these responses to wind loads and water currents, helping to design aerodynamic profiles and damping systems to prevent failures.²⁹³ Extreme water pressure on dams is also assessed with FSI simulations to evaluate structural integrity during events like floods.²⁹⁴ Research on bridge damage from flooding used a volume of fluid method with structural dynamics to analyze fluid–solid interactions.²⁹⁵ Vulnerability analyses of bridge structures under extreme flood or tsunami loads used phase field theory and ALE-based FSI coupling to simulate deformability and wall conditions.²⁹⁶ Other studies included a volume of fluid-discrete element method model for underwater structures,²⁹⁷ and a combined reliability and flow analysis for dam spillways.²⁹⁸ The SPH method was used to examine bridge pier dynamics under dam-break floods^{299,300} and bore-like flows.³⁰¹ Additionally, a solver was developed to model hydraulic bridge failures using FSI,³⁰² and the FEM was

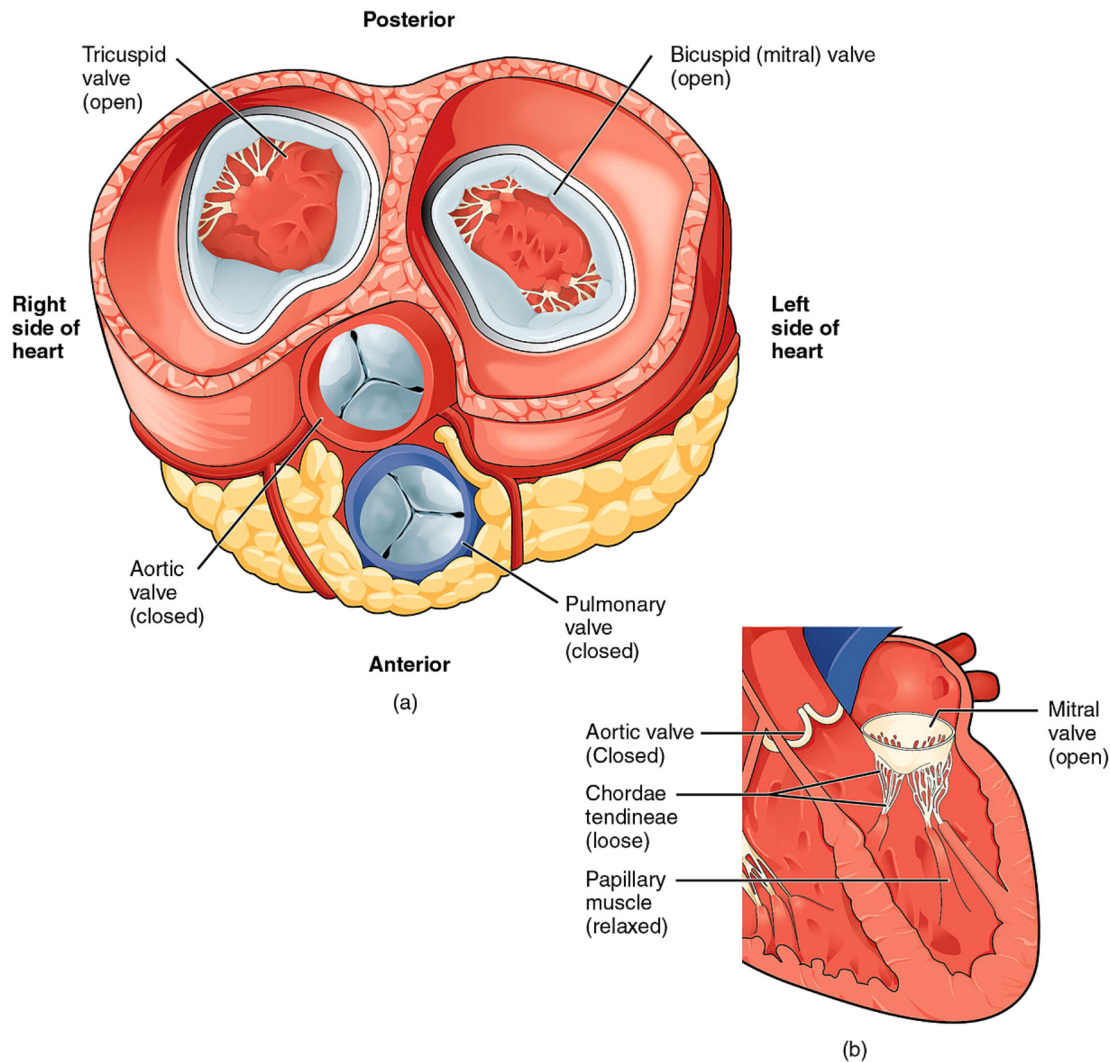


FIG. 13. Schematic representation of the cardiovascular system showing the valves and other components in (a) transverse cross-sectional and (b) coronal planes. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-3.0 license. It is attributed to OpenStax College.

applied to analyze seismic behavior in water storage tanks.³⁰³ Advances in the particle finite element method (PFEM) were proposed for complex fluid–soil–structure interactions,³⁰⁴ and a recent 3D FSI model assessed tsunami impacts on flexible structures.³⁰⁵

Computational FSI is crucial for simulating interactions between fluid flow and pump or turbine components, predicting performance, and optimizing designs. FSI analysis of floating wind turbines using ANSYS explored platform sway and vortex-induced vibrations.³⁰⁶ Impeller FSI studies with CFX and ANSYS Workbench analyzed turbulent flow and structural responses, revealing performance and stability insights.³⁰⁷ A review of FSI in Francis turbines covered flow-induced excitation and fatigue loading,³⁰⁸ while a transient CFD-FEM simulation examined blade stresses.³⁰⁹ Flow conditions in shaft tubular pumps were analyzed using a two-way FSI method, highlighting stress concentrations and pulsation effects.³¹⁰ Impeller vibrations in vertical pumps were studied with one-way FSI,³¹¹ and strength calculations for

horizontal axial-flow pumps used unidirectional FSI.³¹² FSI was applied to high-head pump-turbines and head cover bolts for stress analysis.^{313,314} Thin film FSI in centrifugal pumps was explored using Lagrangian formulations and OpenFOAM.³¹⁵ Bulb turbine wheel dynamics were analyzed with the ADINA software,³¹⁶ and diffuser pump characteristics were studied with two-way FSI coupling.^{317,318} Radial volute pump studies focused on pressure pulsations and hydraulic excitation,³¹⁹ and a review addressed FSI's role in hydrodynamic damping and resonance.³²⁰

III. RELEVANCE OF FSI

A. Biology

Computational FSI in biology and nature involves using computational methods to study how fluids and structures interact in biological systems and natural environments.⁶⁹ This approach helps understand complex phenomena such as blood flow in arteries, the

movement of fish in water, and the bending of plants in the wind, among others.⁶⁴ By modeling these interactions, scientists and researchers can gain insights into the mechanics of life and natural processes, leading to advancements in fields such as biomechanics, ecology, and environmental engineering.³²¹ The boundless occurrences of FSI in nature can never be overlooked. From fluttering of leaves, movement of trees and branches, swimming of aquatic organisms, flying of birds and insects, cytoplasmic streaming and organoid formation, to the respiratory and sensory organs, and cardiovascular systems, the prevalence of FSI in biology is irrefutable to such an extent that this has intrigued researchers ever since.^{322–326} Several instances of FSI in aerial habitats such as plants and birds can be seen in Fig. 14. Not only in plants, birds, and aquatic lifeforms,⁶⁶ FSI is omnipresent inside the human body as well, e.g., in respiratory^{327–332} and cardiovascular systems^{333–336} as well as sensory organs^{337–345} and aneurysms.^{346–350} For instance, FSI has been studied for intricate and statistically defined biological media, e.g., homogenization and a three-dimensional multi-compartmental poro-elastic finite element model has been proposed to understand brain biomechanics that mechanically modeled the fluid transport through the brain parenchyma.³⁵¹

1. Aerial habitats

Computational FSI of aerial habitats involves simulating how fluids (like air) interact with and affect the structures of habitats designed for airborne environments.³⁵² This approach is used to analyze and optimize the performance and stability of these habitats, considering factors such as aerodynamic forces, structural deformations, and environmental conditions.³⁵³ By employing computational models, engineers can predict how these habitats will behave under various conditions, leading to more efficient designs and improved safety in aerial applications.^{354–356} The FSI between airflow and trees (or even

forest stands) includes the reduction in near-surface wind speed as well as generation of turbulence by trees.^{357–359} These are manifestations of a wide range of spatiotemporal scales that further have profound and diverse effects on vegetation.^{360–362} When a fluid flows across a slender soft elastica, the latter deforms so as to experience lesser drag forces.³⁶³ While in the case of static (or quasi-static) flow fields, the structural kinematics are largely unaffected after the transient phase, the situation is quite different when dynamic flow fields are present. For instance, the flow of wind gusts across a tree causes it to bend (or sway) so as to reduce the projected surface area (to reduce drag).³⁶⁴ When the flow field changes, the trees undergo continuous deformation to re-adjust themselves. During this process, the airflow field is disturbed and the wind path also becomes obstructed, leading to strong two-way FSI (often at moderate to high Reynolds numbers).³⁶⁵

The passive flow control enabled by FSI in plants has been studied using micro-networks (microfluidic models). The cell network was linked by nonlinear valves that permitted the physiological internal flows resulting in FSI. A similar phenomenon is observed at a smaller scale when the leaves vibrate in the presence of a moving fluid. Often, they flutter giving rise to instabilities and (atypical) growth phenomenon. Even so, trichomes in plants experience such mechanical interaction with their surroundings. Further down the length scale, the mass transport of food species in vascular plants involves fluid flow across cell walls (xylem/phloem).³⁶⁶ Recently, a numerical model based on large eddy simulation and the immersed boundary method has been proposed to simulate the flow–vegetation interaction between inertial flows and flexible vegetation.⁷⁰ It was observed that alternate vortices (with opposite sense of rotations) appeared at the flow–vegetation interface that resulted in wave-like coherent motion of the vegetation canopy.⁴²

Computational models of FSI simulation for a turbulent flow and flexible rods have been developed using a semi-resolved coupling

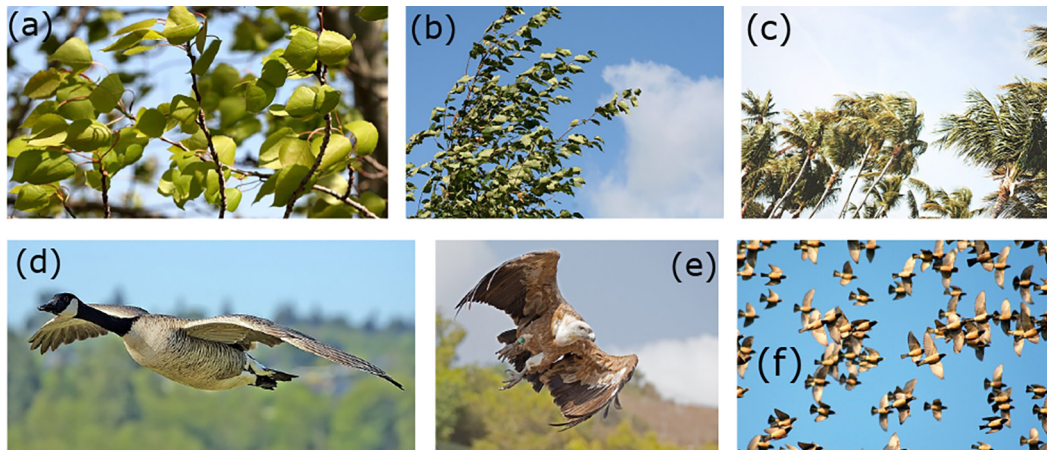


FIG. 14. FSI in aerial habitats such as plants (top) and birds (bottom). (a) and (b) Swaying of leaves in wind. CC-BY-2.0. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-2.0 license. It is attributed to Sharon Mollerus. (c) Palm trees blowing in the wind. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC0 license. It is attributed to Kelley Bozarth. (d) A Canada goose flying at Burnaby Lake Regional Park (Piper Spit), Burnaby, British Columbia. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-2.5 license. It is attributed to Alan D. Wilson. (e) Griffon vulture in flight over Carmel Mount, Israel. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Artemy Voikhansky. (f) Birds—Flock of Rosy Starlings *Pastor roseus*, near Delhi, India. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Thomas Abey.

model promoted by an anisotropic Gaussian kernel function;³⁶⁷ this kernel has been used to account for the influence of the rod with the anisotropic cross section. The finite element method was used to discretize and solve the governing equations of the flexible rod dynamics, while the fluid flow was simulated using the finite volume method. The interaction between fluids and solid rods is modeled by introducing body force terms into the momentum equations. The dynamical behavior and flapping states of clamped flexible fibers immersed within a wall turbulence were predicted using direct numerical simulations;³⁶⁸ it was observed that the clamping to the wall prevented the synchronization of the fiber with turbulent eddies. The collective dynamics and emerging behavior of dense hairy structures within a turbulent flow have been modeled using direct numerical simulations at high Reynolds number.^{369–371}

One of the most energy-efficient lifeforms on earth are the birds and flying insects³⁷² because they flap their airfoil shaped wings in a unique manner for flying through air to minimize the drag and air resistance^{373–375} (see Fig. 15). Specifically, they use both the spatial (similar to ciliary symmetry breaking) and temporal asymmetries (inertia) for aerial locomotion. An immersed-boundary flow solver has been developed along with a nonlinear finite-element solver (for solid mechanics) to study the three-dimensional FSI of elastic insect wings and flow-induced vocal cord vibration. Large deformation geometric nonlinearity as well as material nonlinearity (due to large strains) were accounted for in the solver. The FSI was achieved through a strong interface coupling and partitioned approach.¹⁸³ The aerodynamic force

fields, vortex structures, and associated FSI in fast flapping insect wings and morphing wings in bats and birds have been discussed in a recent review.³⁷⁶

2. Biological cilia

Computational FSI in biological cilia involves using simulations to understand the interactions between the fluid flow and the dynamic mechanical deformation of cilia.³⁷⁷ This is essential for comprehending how cilia function in biological systems. Numerical FSI models simulate the movement of cilia and their interactions with surrounding fluids.³⁷⁸ These models often involve complex FSI due to the flexible nature of cilia and their movement within a fluid environment. The movement of motile cilia, which can beat in coordinated waves, is modeled to study its effects on fluid flow. In the context of the lung and trachea epithelium, the dynamic interaction between the airflow (fluid) and the structural tissues (epithelium and other supporting structures) of the respiratory system gives rise to FSI (see the scanning electron microscope image of lung trachea epithelium in Fig. 16). This helps move mucus and trapped particles out of the respiratory tract. The mucus layer acts as a protective barrier and is crucial for trapping particles. The interaction between the airflow and the mucus affects its distribution and effectiveness in protecting the epithelial cells.³⁷⁹ Furthermore, this knowledge is crucial for diagnosing respiratory conditions, improving treatment methods, and designing effective medical devices.

FSI simulation can reveal how cilia-generated flows affect surrounding fluids, including how they create currents or transport particles.³⁸⁰ This is important for studying both the efficiency of cilia in their roles and the impact on nearby cells and tissues.³⁸¹ In non-motile (primary) cilia, FSI simulations help in understanding their role in sensing and signaling.³⁸² Researchers have often used these simulations to explore how cilia detect mechanical and chemical signals and

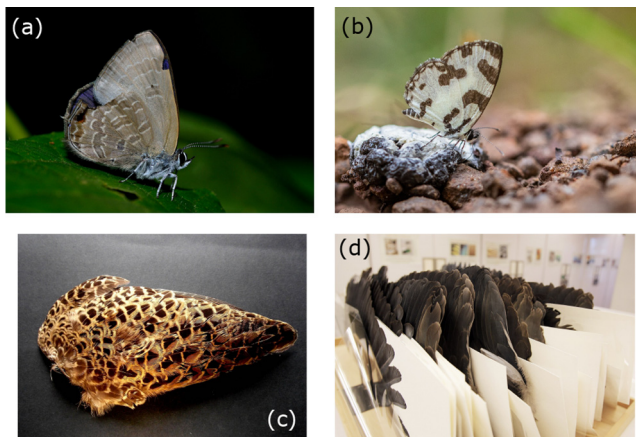


FIG. 15. Birds and flying insects are among the most energy-efficient lifeforms on earth that employ unique wing-flapping techniques with airfoil-shaped wings to minimize drag and air resistance during flight. (a) Close wing position of *Anthera emolus*—common ciliate blue. This picture was taken during the project Wiki Loves Butterfly in Chintamani Kar Bird Sanctuary, Kolkata, West Bengal, India. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-3.0 license. It is attributed to Atudu. (b) *Caleta decidia*—angled Pierrot puddling on bird dropping. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Manaska Mukhopadhyay. (c) From Mal Corvus Witchcraft & Folklore artifact. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-3.0 license. It is attributed to Malcolm Lidbury. (d) Wing specimens in sleeves in Ornithology collection at the Burke Museum of Natural History and Culture. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-3.0 license. It is attributed to Andrea Michelbach.

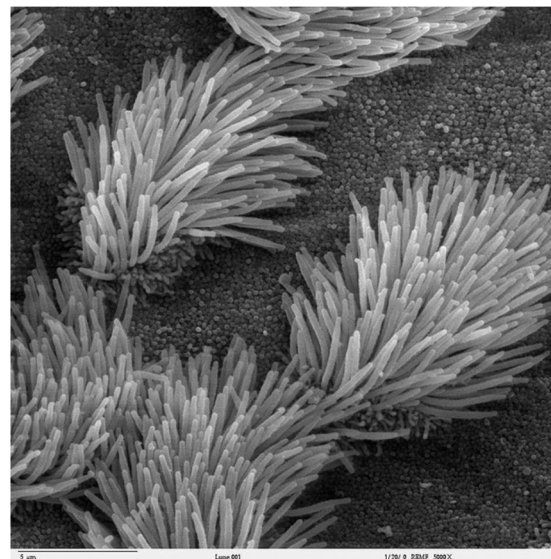


FIG. 16. Scanning electron microscope image of lung trachea epithelium. This work has been released into the public domain by its author, Charles Daghljan. This applies worldwide.

translate them into cellular responses. Computational FSI in biological cilia aids in medical research, such as studying ciliopathies (diseases caused by dysfunctional cilia) and designing better treatments.³⁸³ It contributes to the development of biomimetic devices inspired by ciliary motion in several microorganisms such as single-celled ciliate *Tetrahymena thermophila*, and *Aplysia californica* (see Fig. 17). Overall, computational FSI provides insights into how cilia function in biological systems, enhancing our understanding of their roles in health and disease.³⁸⁴

A plethora of numerical methods could be employed to simulate the complex interactions between fluid dynamics and structural mechanics.³⁸⁵ For instance, FEM could be used for modeling the structural mechanics of cilia, including their deformation and mechanical response to fluid forces, by solving the equations governing the elasticity and movement of cilia.³⁸⁶ On the other hand, FVM can be applied for solving the NS equations in fluid dynamics to simulate the fluid flow around and through cilia. FVM is commonly used in CFD to capture the interactions between the fluid and the ciliary structures.³⁸⁷ Furthermore, IBM can be utilized to handle the interaction between the fluid and flexible structures like cilia.³⁸⁸ It allows the simulation of complex boundary conditions where the fluid and structure interact. Additionally, this method facilitates the simulation of fluid flow

around deforming cilia without requiring a mesh that conforms to the structure's movement.

Mesh-free SPH method can be used for fluid simulation that is particularly useful for complex, dynamic interactions to analyze the fluid dynamics around cilia in certain scenarios where traditional grid-based methods are less effective.³⁹⁰ Even LBM can be used for simulating fluid flow with complex boundary conditions and interactions.³⁹¹ BEM can be applied to model the interaction between the fluid and structure, focusing on the boundaries of the ciliary structures.³⁹² This is useful for problems where the domain is infinite or semi-infinite, such as the fluid environment around cilia. Finally, DNS has the ability to provide high-fidelity results by solving the full NS equations without turbulence modeling for detailed studies of the fluid dynamics around cilia,³⁹³ especially in research requiring precise resolution of flow features.³⁹⁴ These methods can be used individually or in combination, depending on the complexity of the system and the specific aspects of cilia dynamics being studied.³⁹⁵

3. Aquatic lifeforms

Microorganisms use (slender and tubular) ciliary appendages for effective maneuvering through viscous media using synchrony and breaking symmetry involving fully coupled FSI.^{396–400} For example, a chemo-mechanical model of sperm locomotion was developed, showing that periodic flagellar oscillations driven by dynein motors enable movement.⁴⁰¹ The model, incorporating both motor kinetics and fluid mechanics, revealed two distinct swimming modes with different waveforms and velocities. Nonlinear simulations demonstrated that sperm swimming velocity peaks at low sperm numbers and varies non-monotonically with dynein activity, providing insights into swimming efficiency and behavior. The fish species (including cephalopods) in marine environment involve FSI for effective fluidic propulsion.^{402–407} Strong FSI occurs when aquatic lifeforms interact with their surrounding viscous fluid. Interaction between fishes, squids, turtles, and jellyfishes with their viscous fluidic setting has been extensively studied by several researchers over the last decade^{408–410} (see Fig. 18 for a schematic representation of FSI in aquatic lifeforms). For example, the jellyfish, arguably one of the most efficient aquatic species, uses spatiotemporal symmetry-breaking mechanisms for effective underwater propulsion using interactive FSI with the surrounding fluid.^{411–413} Notably, they have suited umbrella-lappet morphologies to adaptively manipulate the surrounding flow field for enhanced fluidic interaction.⁴¹⁴

A dynamic model involving thrust, drag, added mass, and gravity has been used to derive the expression for the instantaneous swimming velocity of jellyfish.⁴¹⁵ The fact that jet propulsion in jellyfish *Aurita* is thrust by jetting, was established in this work. Reynolds number limits for jellyfish propulsion have been estimated using numerical models.⁴¹⁶ This work concluded that paddling or jet propulsion did not work in Stokes flow due to negligible inertia forces. However, these propulsion techniques had pronounced the effect at higher Reynolds number, when inertia forces are in similar orders of magnitude as the viscous forces. Swimming performance and jet-paddling in Rhizostomeae jellyfish *Catostylus mosaicus* has been reported.⁴¹⁷ The functional morphology and fluidic interaction of the Scyphomedusa *Aurelia aurita* jellyfish have been reported,⁴¹⁸ wherein the transformation from discrete lappets in low Reynolds number, to a solid umbrella-like adult form in high Reynolds number has been analyzed

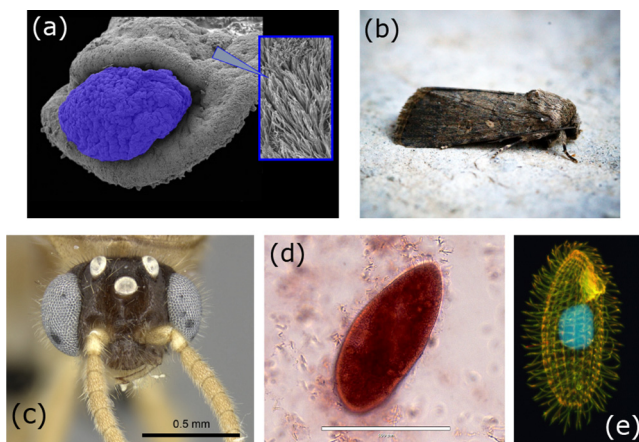


FIG. 17. Ciliated (micro)organisms. (a) Scanning electron micrograph showing a hamster oocyte cumulus complex, colored blue, entering the ostium of an infundibulum. The outer and inner surfaces of the infundibulum are covered with cilia (inset). This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-3.0 license. It is attributed to Prue Talbot and Karen Riveles. (b) Dark Mottled Willow (*Spodoptera ciliata*). This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-2.0 license. It is attributed to Ben Sale. (c) *Adetomyrma ciliata*. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-3.0 license. It is attributed to Ryan Perry. (d) Paramecia showing cilia on cell surface. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC0 license. It is attributed to Fairbanks24. (e) Single-celled ciliate *Tetrahymena thermophila*. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-2.5 license. It is attributed to R. Robinson.³⁸⁹ Copyright © 2006 Public Library of Science. This is an open-access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium.

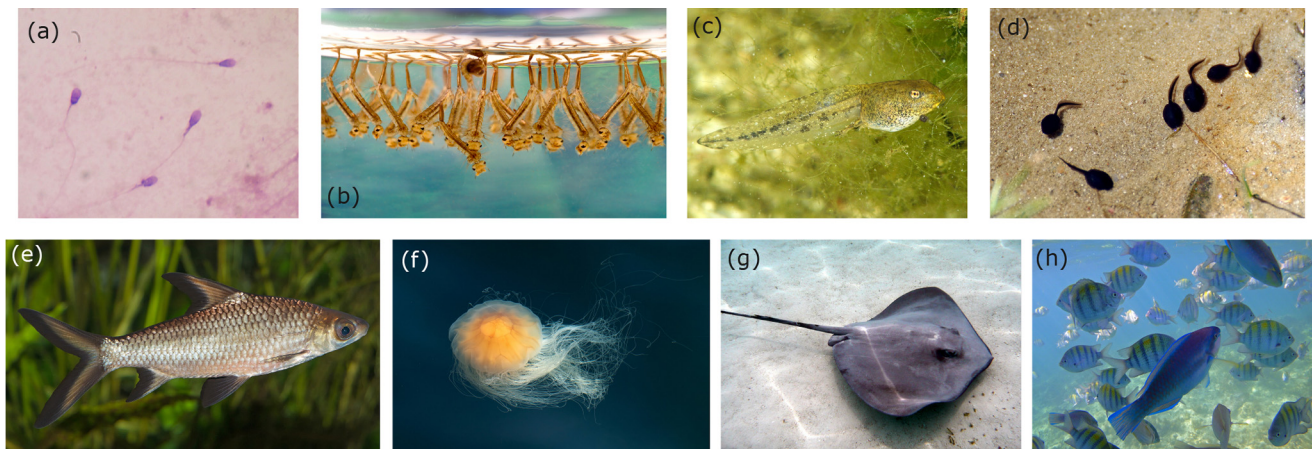


FIG. 18. Examples of aquatic lifeforms that involve FSI. (a) Stained human sperm cells. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Bobigalindo. (b) Larvae of *Culex* Mosquitoes. As seen on the picture, larvae make dense groups in standing water. A shift in the feeding behavior of those mosquitoes helps explain the rising incidence of West Nile virus in North America. It appears that the darker structure at the top center of the image is one pupa. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-2.5 license. It is attributed to James Gathany. (c) Tadpole of the common frog *Rana temporaria* just before metamorphosis. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-3.0 license. It is attributed to Viridiflavus. (d) Tadpoles in a pond near Borzęcin, Lesser Poland Voivodeship. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Jakub Halun. (e) *Balantiocheilos melanopterus*—Bala shark, Karlsruhe Zoo, Karlsruhe, Germany. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-3.0 license. It is attributed to H. Zell. (f) Lion's mane jellyfish in Gullmar fjord at Sämstad, Sweden. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC0 license. It is attributed to W. Carter. (g) Lateral view of a stingray in the water with its stinger in view. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC0 license. It is attributed to Rayjo. (h) Snorkeling off the shore of Punta Cana, Dominican Republic. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-2.0 license. It is attributed to Allen McGregor.

with changing ambient fluid setting. The tunable morphogenesis in phenotypic plasticity facilitated FSI and associated flow dynamics.⁴¹⁹ Advantages of efficient FSI, vortex-shedding, and swimming dynamics of jellyfish over other metazoans have also been discussed.⁴¹⁰

Fish hydrodynamics have been studied using FSI ALE models,⁴²⁰ where the two-way coupled interactions often gave rise to self-sustained oscillations (like the flapping of a flag). The disturbances formed (due to von Kármán vortex) past the fish were calculated to estimate the drag forces upon fish deformation. The FSI solver in OpenFOAM was used for this study, where a steady Poiseuille flow was imposed at the inlet boundary inside a two-dimensional channel flow with traction-free outflow; the flag was modeled as a linearly elastic Hookean solid. Although praiseworthy achievements have been reported in recent years, many challenges still remain in understanding the underlying physical mechanisms of FSI in nature.^{76,421–423} Swimming hydrodynamics and aerodynamics during flying have considerably mesmerized engineers and biologists to concentrate their focus on the emerging field of biomimetics, and soft nature-inspired robotics to match (and even outperform) their natural counterparts.⁷² Undoubtedly, this is an active area of research for its expanding usage as bio-inspired biomimetic soft artificial structures operating inside a fluidic environment.

B. Soft robots

Since the early 21st century, soft robotics has been one of the most intensely and diversely studied fields of scientific research,^{424–426} e.g., in soft matter and materials science communities owing to their

ever-increasing and breakthrough applications in microfluidic devices and biomedical sectors.^{427–429} From targeted drug delivery and laparoscopy to small-scale bio-manipulation and organ-on-a-chip devices,⁴³⁰ miniature soft robots have a plethora of biotechnological and clinical applications that typically involve their use in different forms such as robotic arms, swimmers, and grippers.^{431–433} In most cases, the flexible robotic swimmers have a rich biological inspiration that ranges from small-scale microtubular structures like cilia³⁹⁵ and flagella,⁴³⁴ to spermatozoa⁴³⁵ and jellyfish.⁴³⁶

1. Bionic flaps

The propulsion of actively flapping wings through viscous fluidic environments has been studied to understand the fully coupled two-way FSI that is omnipresent in these compliant systems.⁶³ For example, three-dimensional FSI of elastic beams subjected to transverse steady cross-flow has been discussed,⁴³⁷ where Reynolds averaged Navier–Stokes (RANS) $k - \epsilon$ turbulence model has been used for fluid modeling and the Newmark method oscillator system modeled the structure. In another study, the drag-reduction through reconfiguration has been addressed in the case of flexible beams under cross wind flows, where the end flutter and its instability was studied for varying ranges of wind flow speeds.⁴³⁸ The stability analysis was complemented using reduced-order structural dynamic models. In this regard, see Fig. 19 for a schematic representation of airfoils in nature and various vehicles.

Along these lines, it has been observed that the adaptive reconfiguration of cantilevered elastic blades in steady and oscillatory cross-

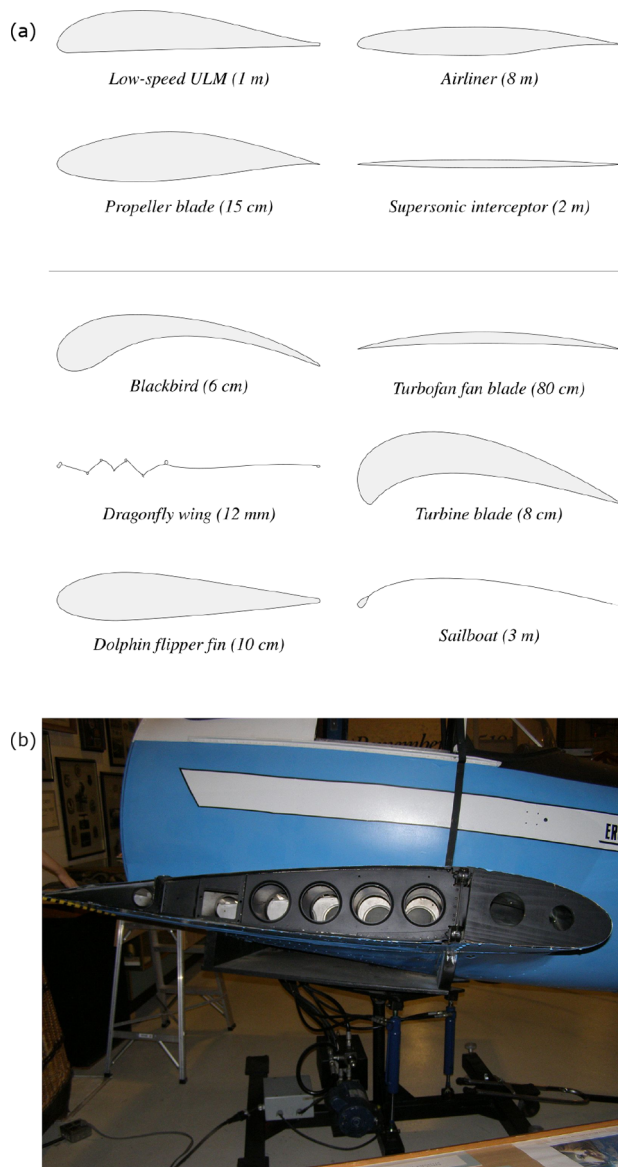


FIG. 19. (a) Selected airfoils in nature and various vehicles, with their approximate chord length indicated. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC0 license. It is attributed to Olivier Cleynen. (b) Airfoil in The Aviation Museum of Kentucky, Lexington. This image has been obtained by the author(s) from the Wikimedia website where it was made available under a CC-BY-SA-4.0 license. It is attributed to Chris Light.

flows significantly alleviated internal stresses.⁴³⁹ The oscillator response has been studied as a function of blade geometry, material properties, and fluid flow fields. Nonlinear coupled elasto-viscous mechanics of soft elastomer cantilever hair beds have been analyzed,⁴⁴⁰ wherein the coefficient of the fluid velocity vs the drag force (Vogel's exponent) has been observed to deviate from the default value (of 2). In this study, the drag-reducing nonlinearity has been discussed for different fluid flow speeds and boundary conditions. The edge flutter

of long beams under follower wind loads has been studied where the flow-induced instabilities (in the form of buckling and flutter in short and long beams, respectively) have been reported.⁴⁴¹ The domains of instability and its correlation with the wind speed have been discussed as well.

Turbulent flows at high Reynolds number past a rotating cylinder have been described in several studies;^{442–444} here, the influence of ratio of angular speed and free-stream velocity upon flow instability has been closely related. Asymmetric von Kármán street has been observed where the asymmetry increased with angular speed. The stream-wise velocity distribution downstream of a cylinder has been reported, alongside understanding the effect of Reynolds number on horse-shoe vortices. 3D separated boundary layer flow has also been noticed besides vortex-shedding and separation line. The flutter of long flexible cylinders in axial flow has been reported previously.⁴⁴⁵ The concept of fluid–elastic instability was initiated in this study that described the stability analysis of very long cylinder edge flutter through a finite difference scheme.^{446–449} Vortex-induced-vibration (VIV) of flexible cylinders bent by transverse flows has been reported by several researchers in the past;^{450–453} these were manifestations of oscillatory fluid dynamic loads on soft elastica.⁴⁵⁴ The wake generation downstream due to adverse pressure gradient and flow separation created an oscillating lift force.⁴⁵⁵ 2D FSI was studied using an immersed boundary technique where the spatial configuration of a filament-like elastica was estimated under incoming wind loads based on their bending compliance⁴⁵⁶ (see Fig. 20). In general, slender cantilever structures have their own natural frequency and there is also a frequency of vortex shedding;^{457–459} resonance occurs when there is a match between the two and Tacoma–Narrows incident is naturally a classic consequence of this phenomena.⁴⁶⁰

Detachment occurs when there is an unsteady flow past a bluff body.⁴⁶¹ Coherent structures develop in the form of von Kármán shedding (global instability), longitudinal vortices (secondary instability), and small-scale Kelvin–Helmholtz vortices (convective instability).⁴⁶² There are nonlinear interaction between the organized, coherent, and turbulent motions.⁴⁶³ The influence of the Strouhal number, vortex-shedding frequency, and resonance on flutter and structural instability has been discussed.⁴⁶⁴ To account for the large-deformation FSI for simulation of a flapping flag, a first-order accurate IBM has been coupled with a structural model to capture the complex flag deformation.⁴⁶⁵ Here, the structural model was based upon a finite difference method. However, in the general framework of solid mechanics, the finite element method is the usual approach undertaken to study nonlinear structures and their intricate deformation and complex material behavior. Nowadays, this method combines with IBM, thus paving the path for large-deformation FSI applications. A study has reportedly done so—however, the study was limited to two-dimensions and linear material behavior.⁴⁶⁶

Laminar flow past a vibrating beam attached to the lee side of a circular cylinder has been proposed.⁴⁶⁷ 3D turbulent near-wake structure in the case of a flat plate during high Reynolds number has been experimentally studied using tomographic particle image velocimetry (PIV).⁴⁶⁸ In this study, flow around a thin flat plate with sharp edges was studied for its 3D velocity field. Strongly detached unsteady flow was observed through tomo-PIV database. Reynolds averaging, phase-averaged dynamics, and proper orthogonal decomposition were used for characterizing the flow field. The unsteady fluid transport

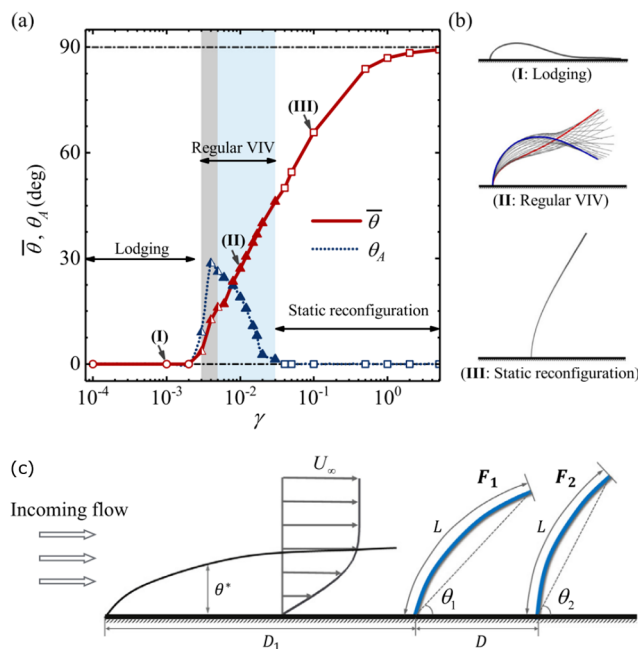


FIG. 20. 2D FSI between flexible filaments subjected to incoming laminar boundary layer wind flows.⁴⁵⁸ (a) Variation of mean inclination angle $\bar{\theta}$ and amplitude of angular oscillation θ_A for different bending stiffness of the single filament interacting with wind. (b) Three distinct configurations of the flexible filament: (I) lodging, (II) regular VIV, and (III) static. (c) Interaction between an incoming flow and the two filaments that deform differently owing to distinct fluidic interaction and their own horizontal spacing; here, vortex fields and streamlines have been reported for this dual-filament system. [Reproduced with permission from Zhang *et al.*, J. Fluids Struct. **92**, 102787 (2020). Copyright 2020 Elsevier].

phenomenon that occurs during fluid–structure interaction and in unsteady recirculating flows was described using Lagrange coherent structures.⁴⁶⁹ In this regard, an excellent understanding of VIV in circular cylinder through modeling and experiments has been reviewed.⁴⁷⁰

In a recent study, an insect-inspired flapping wings have been developed and a dynamic similarity framework has been proposed to understand the strongly coupled FSI involved.⁴⁷¹ A passively deployable flap (equivalent to covert feathers) has been numerically investigated using a projection-based immersed boundary method for its lift enhancement, vortex-shedding, and temporal synchronization of its oscillations at a low Reynolds number.⁴⁷² The generation of lift-based thrust in flapping foils has been studied using the Knoller-Betz effect, where the thrust is essentially the force component that is projected onto the direction of propulsion. The estimation of propulsion performance based on FSI solvers for a bio-inspired nano rotor base has been reported and validated using experiments.⁴⁷³ The influence of FSI upon the aerodynamic performance, force, and efficiency of flapping wings has been computationally studied for a hovering hawkmoth simulated under virtual vacuum conditions.⁴⁷⁴ FSI hydrodynamics along with lift and thrust measurements have been reported in the case of a bionic flipper with non-uniform material stiffness distributions.⁴⁷⁵ Recently, the unsteady flow control of a bird-inspired flexible flap has studied using a combination of the immersed

boundary, finite element, and lattice Boltzmann methods.⁴⁷⁶ Here, the mechanisms of flow separation on the airfoil surface and the vortex generation at the edges have been discussed under different flexible flap positions and varying material (elastic) properties.

2. Microfluidic devices

Some of the important microfluidic, organ-on-a-chip, and biomedical applications that require a fundamental understanding of FSI in soft active elastica are targeted drug-delivery, cell labeling, micro-object manipulation, and laparoscopy.^{479–484} In these examples, tiny machines are often empowered with highly versatile transduction features (with enhanced sensing and actuation modalities⁴⁸⁵) adaptive and manipulative FSI (with the surrounding fluid) for applications such as cargo vehicles and lab-on-a-chip devices.^{486–488} Nature-inspired stimuli-responsive soft materials (that classify themselves as active elastica) that typically operate inside a fluid medium experiences drag forces offered by its surrounding viscous environment.^{489–494} Therefore, in dynamic force field conditions and time-dependent systems, FSI becomes relevant (see Figs. 21 and 22).^{477,495–498} A critical review of fabrication methodologies, actuation strategies, motion kinematics (including particle transport phenomena and flow generation/sensing), and state-of-the-art microfluidic applications of artificial magneto-responsive cilia has been reported.⁴⁹⁹

Inspired by naturally occurring ciliates, arrays of cilia have been fabricated to generate programmable nonreciprocal motion and metachronal coordination behavior (see Fig. 23).^{477,500–503} Net fluid propulsion and particle manipulation have been achieved using surfaces partially covered with untethered artificial cilia for anti-fouling and self-cleaning.⁵⁰⁴ The particles were transported by the deformable cilia on their tips along the desired directions with complete external control and precision. The ciliated surfaces were sometimes used for microparticle removal as well.⁵⁰⁵ Partially active magnetic cilia and their arrays have been studied to enable net fluid transport and demonstrate vortex structures (see Fig. 24).⁵⁰⁶ FSI has a significant influence in microfluidic devices that often involve the use of soft hair-like cilia and flagella within a viscous fluidic setting.⁵⁰⁷ Computational modeling of FSI in mucus layers of the respiratory tract has been studied [see Fig. 25(a)].⁵⁰⁸ Here, the ciliary structures have been observed to undergo complex wave motions to push the mucus along a specific direction utilizing non-reciprocal strokes during one motion cycle [similar to the motion of a classic cilium, see Fig. 25(b)].

Magnetically actuated artificial slender microtubules have been developed for microfluidic applications and fluid transport at low Reynolds number flows where polymer films impregnated with magnetic nanoparticles have been modeled as cilia.⁵⁰⁹ Here, the coupled nonlinear partial differential equations have been solved using Navier (for solid), Maxwell (for magnetic), and NS (for fluid) equations.^{226,478,510} Versatile microfluidic flow has been achieved by molded artificial cilia, similar to flexible hair-like micro-actuator-like biological cilia.⁵¹¹ Collective wavelike metachronal actuation of microscopic artificial cilia has been shown to demonstrate microfluidic pumping.⁵¹² In another study, suspended pillars were used as a stirring system for controlled cell clustering.⁵¹³ Generally, these soft elastica are used for micro-manipulation and micro-mixing. The interplay of magnetic, viscous, inertia, and elastic forces affects their effective FSI and kinematics.⁵¹⁴ In this regard, the influence of spatial, temporal, and orientational asymmetry³⁹⁵ and the effects of Reynolds number

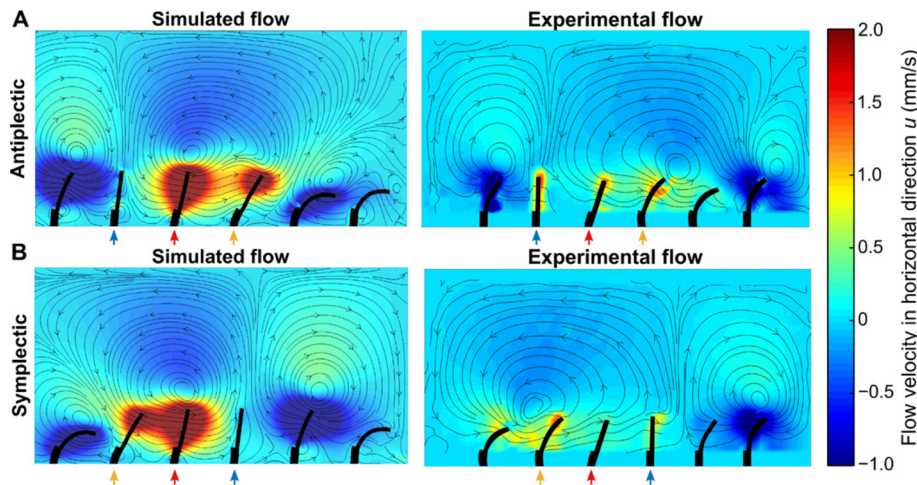


FIG. 21. Numerical simulations (left) and experimental observations (right) of the horizontal fluid velocity fields for *antiplectic* [panel (a)] and *symplectic* (panel B) coordinated ciliary wave patterns with the same phase shift ($-\pi/3$); the third cilium from the left is at the same configuration during the effective stroke. In the case of the symplectic stroke, the fourth cilium is at the end of the recovery stroke; hence, it obstructs the flow induced by the third cilium and induces a vortex. However, in the antiplectic case, the fourth cilium is actually more bent to the right and is also in the effective stroke; therefore, this leads to an enhanced flow.⁴⁷⁷ [Reproduced with permission from Milana *et al.*, *Sci. Adv.* **6**, 49, eabd2508 (2024). Copyright 2024 American Association for the Advancement of Science].

and corresponding fluid inertia have been discussed.⁴⁸¹ For microfluidic applications, rotational flows are sometimes required and this has been realized using remotely actuated magnetic rotating microbots that induce net microfluidic flows even at low Reynolds number.⁵¹⁵

Recently, a multiscale biophysical model revealed quantized metachronal waves in ciliary lattices that focused on how ciliary bed morphology, beating patterns, and interactions influence wave dynamics.⁵¹⁶ By incorporating detailed ciliary mechanics, the model elucidated mechanisms behind metachronal wave emergence and self-organization across scales from nanometers to millimeters,

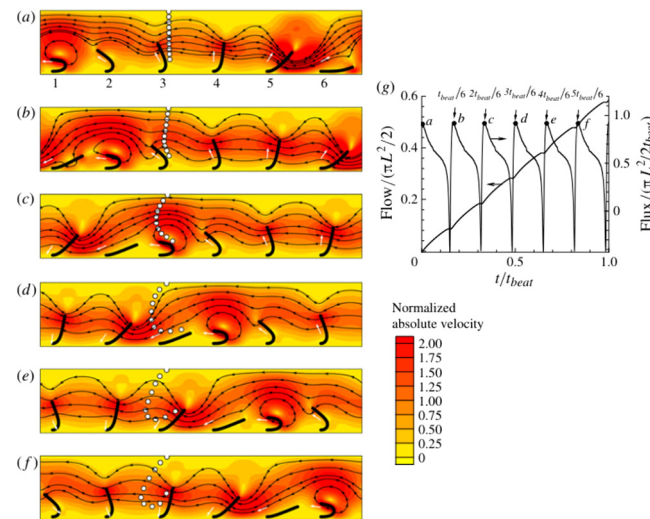


FIG. 22. (a)–(f) FSI between an array of magnetically actuated cilia with the surrounding fluid.⁴⁷⁸ Out-of-phase motion of cilia during a representative cycle. While the net fluidic transportation is to the left, the ciliary waves travel to the right (antiplectic). The contours represent normalized velocity, and the streamlines represent the flow direction. The magnetic field is shown by white arrows. The instantaneous flux and flow with increasing time is shown on the right. Subsequent time steps are marked as (a) $t = 0$, (b) $t = t_{beat}/6$, (c) $t = 2t_{beat}/6$, (d) $t = 3t_{beat}/6$, (e) $t = 4t_{beat}/6$, and (f) $t = 5t_{beat}/6$. [Reproduced with permission from Khaderi *et al.*, *J. Fluid Mech.* **688**, 44–65 (2011). Copyright 2011 Cambridge University Press].

highlighting potential applications in developmental biology and biomedical fields. Another study explored spontaneous oscillations and hydrodynamics in active microfilaments, focusing on the axoneme's structure and dynein motors.⁵¹⁷ By extending a sliding-controlled axonemal feedback model, the research derived nonlinear governing equations incorporating biochemical noise and long-range hydrodynamic interactions. The model predicted wave transitions, including reversal in propagation direction and varied waveforms akin to spermatozoa beats. It also accounted for asymmetric ciliary beats and generalized dynein regulation, providing insights into flagellar dynamics and minimal models for swimming microorganisms.

The hydrodynamic synchronization in beating filaments has been investigated using a geometric feedback model of the axoneme.⁵¹⁸ It found that both in-phase and anti-phase synchrony can occur with asymmetric beats, while symmetric waveforms synchronize in-phase. The research also revealed phase slips caused by biochemical noise and highlighted the roles of hydrodynamics and mechanochemical feedback in synchronization, aligning with recent experimental findings. In another study, the cytoplasmic streaming in fruit fly egg cells was analyzed using an active carpet theory.⁵¹⁹ It identified key factors driving intracellular flow dynamics, showing that a complex system simplifies to a low-dimensional attractor for swirling flows. The findings provide insights into universal aspects of fluid-mediated dynamics, with implications for cellular transport in large cells. Another recent study examined fluid circulation driven by metachronal waves in *T. acetii* nematodes, revealing that their collective movement within fluid drops generates a fluidic circulation (whirling).⁵²⁰ A quasi-two-dimensional hydrodynamics model, using Stokes flow approximation, showed that the nematodes' periodic motion drives the flow, with large tail excursions producing circulation. The findings suggest design constraints for enhancing fluid motion in flow-generating systems.

3. Artificial swimmers

In general, soft robotic swimmers are composed of soft pneumatic composites,¹⁰² ionic-polymer metal composites,^{521,522} nanocomposite hydrogels,⁵²³ elastomers,⁵²⁴ electro-active polymers,^{525,526} shape memory polymers,^{414,527} or soft magnetic elastomeric composites.⁵²⁸ Recently, an optically powered miniaturized soft robotic swimmer has

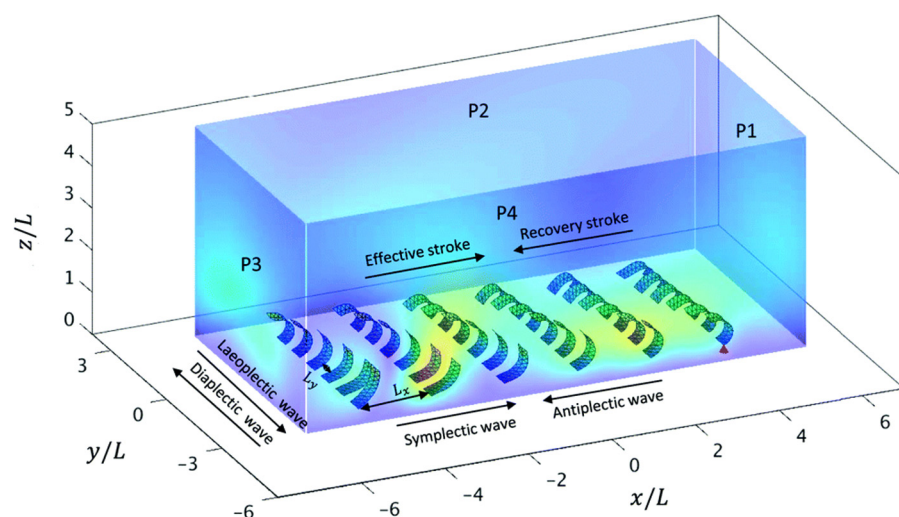


FIG. 23. Numerical simulations of FSI between periodically arranged six-by-six cilia show coordinated wave movements for optimal fluid transport.⁵⁰⁰ [Reproduced with permission from Zhang *et al.*, *Soft Matter* **18**, 20, 3902–3909 (2022). Copyright 2022 Royal Society of Chemistry].

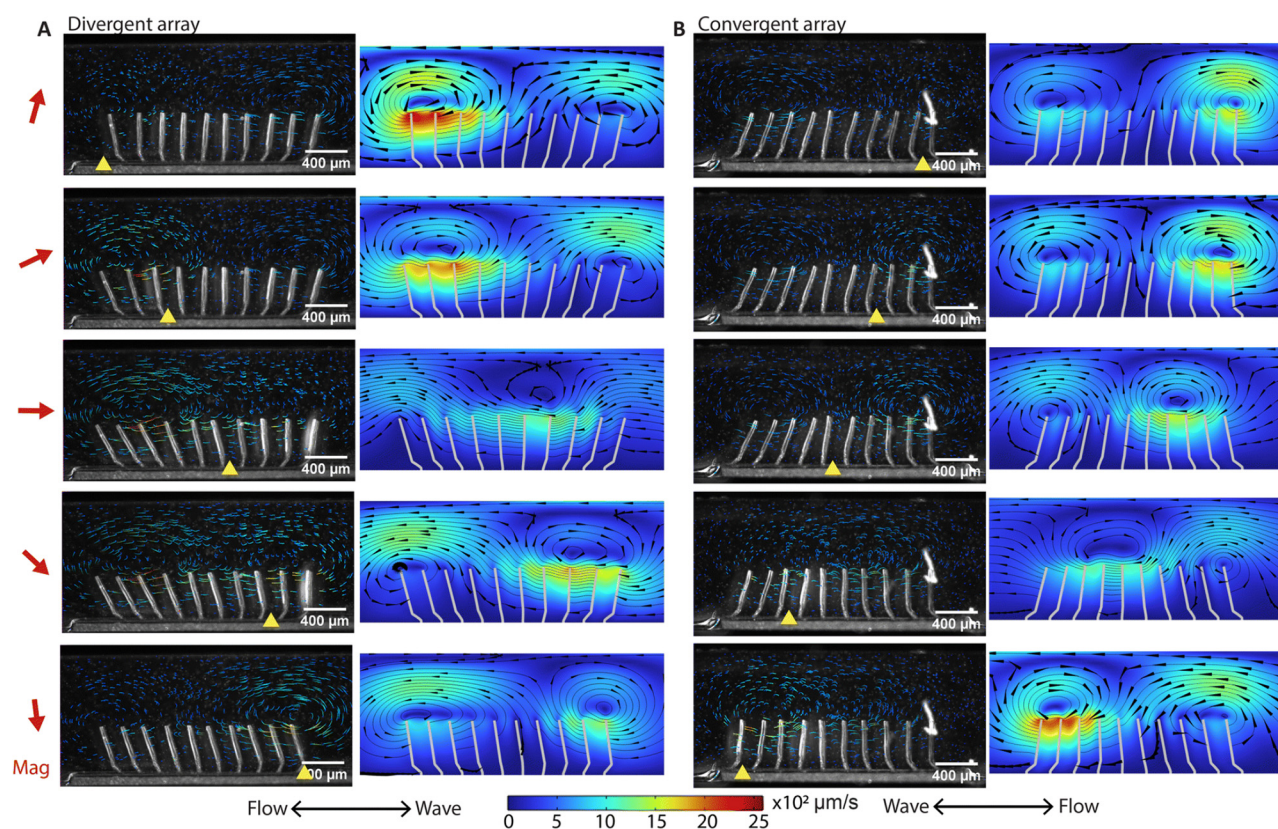


FIG. 24. Metachronal waves and the resulting velocity field due to an array of partially active magnetic cilia. (a) and (b) The experimental results (left column) and simulations (right column) show metachronal motion and the surrounding vortices for the divergent array (a) and the convergent array (b) configurations. The corresponding magnetic field is represented by the red arrow on the left side of the figure. The yellow triangle denotes the position where the cilia engage in the elastic stroke, after which they cannot follow the magnetic field, and therefore, start deforming in the opposite direction. Tracer particles have been used during the experiments to visualize the velocity field. The tail-like lines in the experimental images represent the trajectories of the tracer particles, and the color indicates the magnitude of velocity. In the simulation results, the instantaneous velocity field is shown by the streamlines where the arrows indicate the velocity direction and the color is the magnitude.⁵⁰⁶ [Reproduced with permission from Wang *et al.*, *Lab Chip* **24**, 1573 (2024). Copyright 2024 Royal Society of Chemistry].

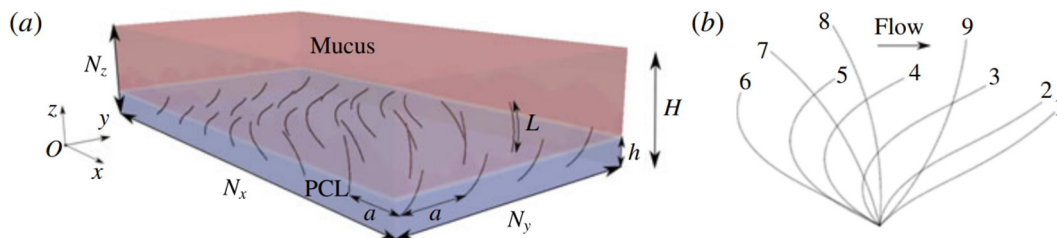


FIG. 25. (a) Computational modeling of the FSI between the periciliary layer (blue) and mucus (red).⁵⁰⁸ Filament-like slender ciliary structures undergo anti-plectic metachronal wave patterns (a specific form of coordinated movement pattern) within a Newtonian fluid domain. Here, a lattice Boltzmann method has been employed to study these complex 3D FSI responsible for the efficient transport of mucus. (b) Ciliary beating pattern with steps 1–6 corresponding to the recovery phase and steps 7–9 to the stroke phase. [Reproduced with permission from Chateau *et al.*, *J. Fluid Mech.* **824**, 931–961 (2017). Copyright 2017 Cambridge University Press].

been fabricated using (opto-genetically modified) biological matter to successfully navigate through a viscous fluid (see Fig. 26).⁵²⁹

These flexible robots are innately made of polymers that are compliant, and when these are embedded with magneto-responsive microparticles in the form of external filler materials, they ensure a pre-defined precise untethered actuation. As a result, they become capable of remote-control, system monitoring, and external actuation through the impregnation of permanently magnetic materials, rendering them programmable shape-morphing features as well.⁵³⁰ Considering their practical applications, these soft robots are often required to locomote through viscous fluids; this is when the compliant robotic structure interacts in a strongly coupled manner with the surrounding fluid, thereby giving rise to fully coupled nonlinear FSI. Further, an intricate interplay develops among the elastic, inertia, viscous, and magnetic forces, which essentially determines their swimming kinematics.⁵³¹ A recent study developed an FSI framework by coupling an FEM-based structural solver with a lumped vortex method-based potential flow solver to analyze fish swimming dynamics. The caudal fin of a carangiform fish was modeled as a cantilever beam with periodic motion, and the effect of actuation frequency on thrust was investigated. The entire fish body, modeled as a free-free beam, included time-dependent muscle moments along its length to capture full-body undulations.⁴⁰⁶

Researchers have reported bio-mimetic design strategies and fabricated such soft robotic swimmers in the form of fishes,⁵³² turtles,⁵³³ humans,⁵³⁴ jellyfishes,⁵³⁵ squids,⁸⁸ snake,⁵³⁶ and flagella.⁵³⁷ These aquatic soft robots exhibit intricate swimming dynamics that feature

characteristically at different length scales; however, their propulsion through viscous fluidic channels is often barbed with intricate multi-physical phenomena.^{538,539} A survey has been carried out that reports the recent developments and motion control strategies in biomimetic underwater robots.⁵⁴⁰ Owing to their criticality and fully coupled nature involving multiple energy domains, these applications require a rigorous understanding of their large deformation FSI.

Several studies have focused on understanding the swimming kinematics and FSI in stimuli-responsive soft robotic swimmers inspired from turtles, octopus, stingray (see Fig. 30), fish (see Figs. 27, 28, and 31), and jellyfish (see Fig. 29). For example, a control model to account for the switching mechanism between planar and helical swimming flagella has been reported for a soft robotic sperm.⁸¹ A computational fluid dynamics-based multi-objective control optimization has been reported for a soft fish-inspired robot incorporating a muscle-like actuation.⁴⁹¹ The computational design of a compliant bionic robotic fish capable of high-speed locomotion has been discussed along with their FSI.⁵⁴¹ In another study, the multi-field coupling behavior has been analyzed and designed for a fish-inspired soft robotic swimmer.⁵⁴² The swimming dynamics of magnetically actuated artificial flagella has also been studied as a fully coupled problem involving magnetics, solid mechanics, fluid

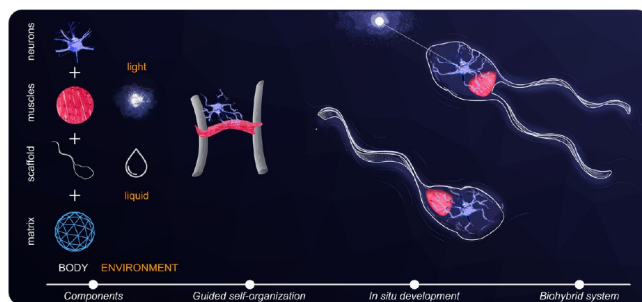


FIG. 26. Schematic representation of an optically powered motile bot (fabricated using optogenetic motor neurons, extracellular matrix, scaffold, and muscle tissues) operating inside a fluidic surrounding.⁵²⁹ [Reproduced with permission from Aydin *et al.*, *Proc. Natl. Acad. Sci.* **116**, 40, 19841–19847 (2019). Copyright 2019 National Academy of Sciences].

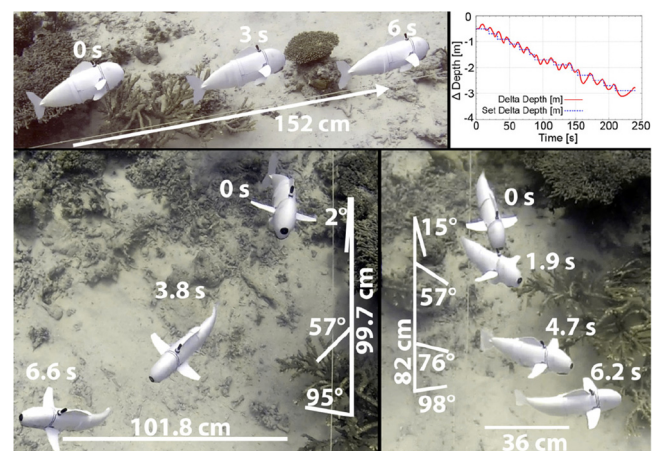


FIG. 27. Acoustically driven soft robotic fish within ocean demonstrates straight swimming and maneuverability.⁷⁸ [Reproduced with permission from Katzschmann *et al.*, *Sci. Robot.* **3**, 16, eaar3449 (2018). Copyright 2018 American Association for the Advancement of Science].

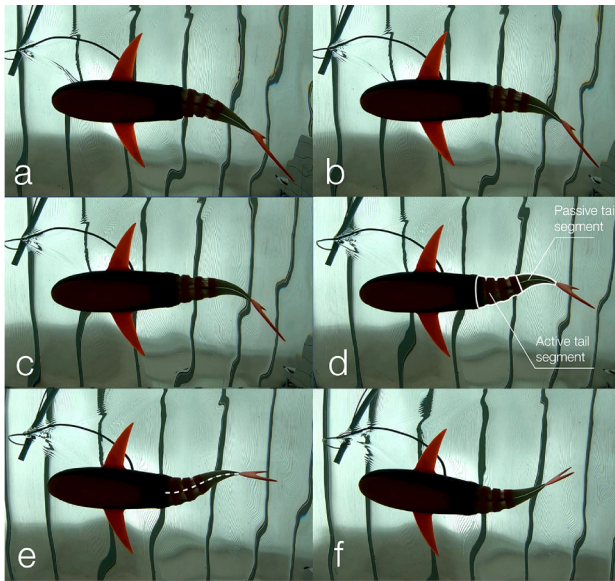


FIG. 28. Chronologically ordered snapshots from a to f of thunniform-like swimming motion of a soft robotic fish realized through a combination of an active and a passive tail segment.²⁶ [Reproduced with permission from van den Berg *et al.*, *HardwareX* **12**, e00320 (2022). Copyright 2022 Elsevier].

dynamics, and large deformation FSI.⁴³⁴ To ensure bio-inspired bacterial swimming and bi-directional motion, coupled FSI numerical modeling based on finite elements and NS equations along with Robin transmissions has been conducted as well.⁵⁴³

Mechanics-based models have been proposed for locomotion of sandfish to understand their high-performance sub-surface sand-swimming.⁵⁴⁴ FSI modeling was performed for a soft robotic fish made of piezoelectric fiber composite.⁴⁹⁰ Inspired by microtubular flagella, miniature swimming robots with multiple such assemblages of flexible flagella have been reported.⁵⁴⁵ A turtle-like swimming robot

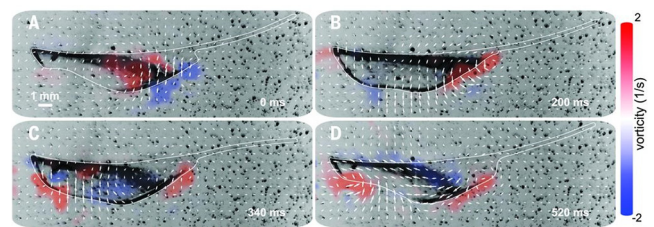


FIG. 30. Particle image velocimetry flow field measurements show alternated (positive and negative) vortices around the soft robotic ray³⁰ for chronological time instants: (A) $t = 0$ ms, (B) $t = 200$ ms, (C) $t = 340$ ms, and (D) $t = 520$ ms. [Reproduced with permission from Park *et al.*, *Science* **353**, 6295, 158–162 (2016). Copyright 2016 American Association for the Advancement of Science].

has been fabricated using a smart composite in the form of a compliant structure.⁵⁴⁶ A similar study on the fabrication of a bio-mimetic robotic jellyfish using shape memory alloy springs has been presented.⁵⁴⁷ Using a coupled FSI model, the dynamics of a soft robotic fish has also been reported along with parametric optimization.⁵⁴⁸ A simplified model has been proposed to account for the mechanical response of a soft hydraulic robot driven by artificial muscle under electro-mechanical actuation.⁵⁴⁹ A thorough analysis involving design, modeling, and experimental characterization was reported for a single-actuated robotic swimmer.⁵⁵⁰ Another electro-mechanics-based modeling and parameter identification has been reported for smart robotic manipulators made of soft electro-active polymers.⁵⁵¹

Computational modeling of swimming kinematics in marine invertebrates for designing, modeling, and controlling the dynamics of soft swimming robots has been reported.⁵⁵² A study on the modeling of compliant robots actuated by twisted-and-coiled actuators has been discussed.⁵⁵³ Miniature flexible robots with three-dimensional remnant magnetization and motion kinematics through pre-programmable structural deformations have been analyzed.⁵⁵⁴ Theoretical and computational studies have been carried out for five-ray flexible pectoral fin during their labriform swimming.⁵⁵⁵ A comparison had been made upon the variation in experimental and

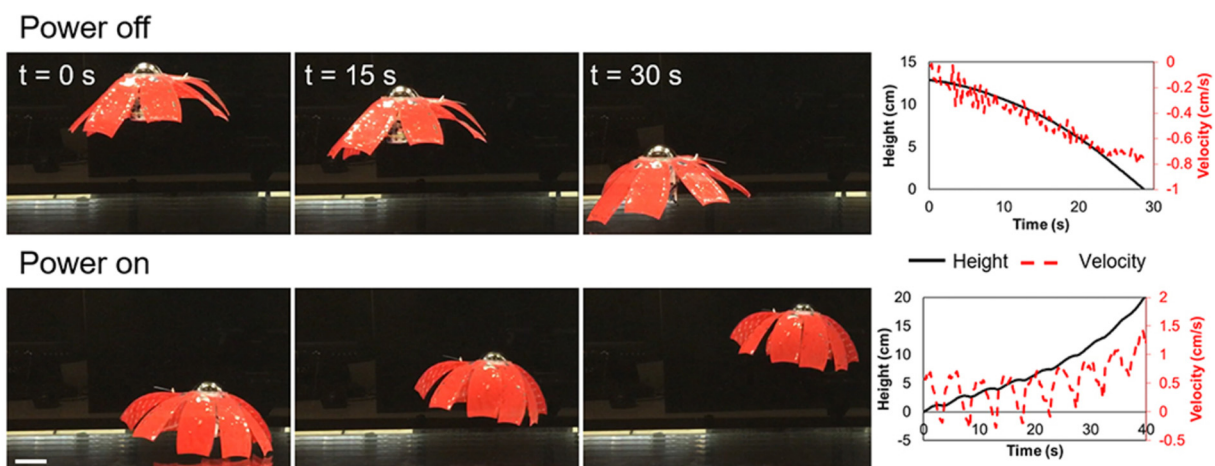


FIG. 29. Electrically driven fluid electrode powered jellyfish-inspired soft robotic fish demonstrates free swimming.⁸² [Reproduced with permission from Christianson *et al.*, *Front. Robot. AI* **6**, 126 (2019). Copyright 2019 Frontiers Media SA].

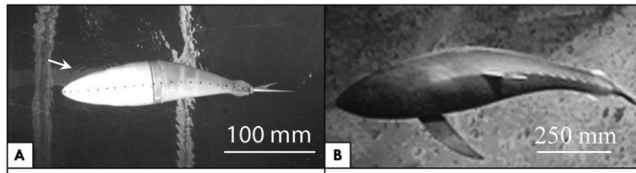


FIG. 31. A tunabot (a) and a yellowfin tuna (B).⁵⁷⁹ [Reproduced with permission from Zhu *et al.*, *Sci. Robot.* 4, 34, eaax4615 (2019). Copyright 2019 American Association for the Advancement of Science].

numerical results for helical swimmers through viscous channel flows.⁵⁵⁶ FSI model of soft robotics using an active strain approach has been proposed.⁵⁵⁷ The dynamics of a viscous liquid within a closed elastic cylinder subject to external forces with application to soft robotics has been explained in another study.⁵⁵⁸ The FSI of a soft robotic swimmer has been modeled using a fictitious domain approach and active strain technique;⁵⁵⁹ here, multiplicative decomposition of the deformation gradient was carried out for Lagrangian finite element structure sub-problem solver, along with a modified NS equation with Lagrangian multiplier for the fluid sub-problem.

Simulation and computational design of soft robotic swimmers with artificial muscle have been discussed.⁵⁶⁰ The computational design of a carangiform robotic swimmer traversing through a multi-physics simulation environment has been presented.⁵⁶¹ A computational fluid dynamics-based analysis and design of an ostraciiform swimming robot has also been reported by the same research group.⁵⁶² Three-dimensional modeling and numerical simulations of multi-modal swimming have been presented for a fin-actuated soft robotic fish.⁵⁶³ Researchers have developed a snake-like swimming robot using actuators made of ionic polymer-metal composites.⁵⁶⁴ High-fidelity numerical simulations have been performed for a bionic robot fish based on macro-fiber composites.⁵⁶⁵ Recently, bio-inspired soft robots with flapping appendages have been designed for enhanced kinematic performance using the model of mobile multibody systems, floating frame approach, and the geometrically exact approach.⁶⁵ The coupled interaction between inertia, viscosity, and elasticity in soft robotic actuators with fluidic network has been discussed.⁵⁶⁶ The hydrodynamic modeling and performance analysis of bio-inspired swimmers have been reported.⁵⁶⁷

Since jellyfish requires the lowest cost-of-transport during their locomotion, it has been a strong motivation for several researchers to develop magnetically actuated jellyfish-inspired soft robotic swimmers.⁵³⁰ Soft robots have been conceptualized from jellyfish swimming dynamics and driven by fluid electrode dielectrics.⁸² An experimental attempt on the jellyfish-inspired soft robot actuated by soft pneumatic composite has been successful in this regard.¹⁰² Biomimetic jellyfish actuated by shape memory composite actuators have been fabricated and reported as well in another study.⁴¹⁴ A 3D printed actuator forming the basis of a jellyfish-like soft robot was fabricated by curing a newly synthesized particle double network (PDN) hydrogel, which contracts in a similar way to the muscles of a moon jellyfish.⁵⁶⁸ A shape-programmable magnetic soft matter was developed with complex geometry mimicking soft jellyfish under external magnetic actuation.⁵⁶⁹ In another group, a soft ferromagnetic continuum robot with hydrogel skin has been fabricated for biomedical

applications, in particular, for navigating and accessing tortuous cerebro-vascular path for minimally invasive surgery.⁵⁷⁰

Based on the architecture and capability of large body deformations of an aquatic octopus, researchers have fabricated and emulated their swimming kinematics to develop self-powered untethered octopus-inspired soft robotic swimmers (octobots) for minimal power loss.^{571–575} Squishy membranes in the form of webbed arms have often been used by octobots to swim efficiently.⁵⁷⁶ Pneumatic networks have also been embedded in the octobot soft body made of silicone elastomeric rubber in order to power it for high swimming speeds.⁵⁷⁷ Tuna and other scombrid fish exhibit a high-performance aquatic swimming when they escape from predators or catch prey.⁵⁷⁸ A biomimetic design and fabrication of a tuna-inspired underwater soft robotic swimmer has been reported and quantified for its swimming performance and surrounding flow field evolution generated by the tail;⁵⁷⁹ this study has explored a fish-like performance space for futuristic bio-inspired underwater vehicles. Similarly, the hydrodynamics of carangiform swimmers and their dependence on undulation wavelengths have been studied through numerical experiments for a model Jack fish swimmer robot.⁵⁸⁰ A biomimetic tuna-inspired robot for underwater exploration has been developed to understand the influence of the caudal fin and its shape upon the propulsive performance.⁵⁸¹

Flexible robotic fish that is capable of large structural deformations during underwater locomotion has been shown to achieve a better swimming performance compared to their rigid-bodied counterparts.⁵⁸² The swimming behavior of such a compliant fish-inspired robot has been studied by combining the large-amplitude elongated-body theory along with the decoupled natural orthogonal complement matrices. Hydrodynamics of rapid linear acceleration that resulted in a thrust-based swimming in tunabots have been discussed.⁵⁸³ Here, experiments were performed to study the tunabot motion kinematics, fluid flow behavior, and mechanical power requirements during unsteady locomotion. The locomotion and swimming kinematics of an autonomous fish-like three-dimensional compliant robot having multiple fins have also been studied.⁵⁸⁴ Bio-inspired swimming robot mimicking a stingray and fabricated using a micro-fabricated gold skeleton and an elastomer body has been reported;⁸⁰ this soft robot was powered by rat-derived heart muscle cells (cardiomyocytes), which are genetically engineered to respond to light sources. Specifically, this was done to generate the required undulation for soft robotic propulsion.

Modeling and design of a swimming magnetic sheet for untethered miniature soft robots have been attempted.⁵⁸⁵ Furthermore, soft robotic swimmers have been modeled using the discrete elastic rod method.⁵⁸⁶ Inspired by lateral undulation, magnetically actuated soft robot with triangular head-tail morphology has been reported as well.¹⁰⁸ Another research group has developed a detailed design, modeling, and characterization technique for a miniaturized compliant robotic swimmer with two vibration modes.⁵⁸⁷ In another study, a bio-inspired robotic fish that was driven by compliant parts has been modeled.⁵⁸⁸ A study on soft continuum robots has been proposed using a geometrically exact model in the framework of the finite element deformation space formulation.⁵⁸⁹ A concise review on the design, modeling, and control of biomimetic fish robots has been presented.⁵⁹⁰ The directed motion and collective behavior of robotic spermatozoa have been reported that deal with the self-propelling motion of stimuli-sensitive hydrogel microswimmers with complex behaviors in

capsules.^{591–593} In conclusion, several studies concerning the kinematic modeling, performance optimization, and actuation/control of nature-inspired miniaturized soft robotic swimmers have been reported using different numerical approaches.

IV. SUMMARY AND PERSPECTIVES

In this review, we have discussed the myriad occurrences (and relevance) of FSI not only in nature and biological systems but also in artificial structures (Sec. I). Their pertinence in soft active and passive elastica has been highlighted. The fundamental physical interaction between these geometrically nonlinear soft structures and their surrounding flow field gives rise to large deformation FSI. Indisputably, these interactions are not only limited to animal and aquatic lifeforms but also found in plants and trees. Even in soft robotic swimmers and microfluidic devices, their presence is beyond doubt. Therefore, several biological and bio-inspired systems involve large deformation of flexible structures under fluid tractions (and external actuation forces) giving rise to these mechanical interactions. Flapping of aerofoils, insect wings, periodic movement of heart valves, and vibration of human vocal folds are only a few examples, wherein the structural domain deforms to sway the surrounding fluid resulting in complex flow patterns and structural motion.

Over the years, a plethora of numerical techniques and computational models have been proposed to study FSI. In Sec. II, we have provided a systematic description of the classifications of FSI and also highlighted the early developments in this field. Often, these FSIs cause excessive mesh distortion with subsequent degradation in the mesh quality during numerical modeling. This gives rise to numerical oscillations and associated numerical errors (mesh-related problems). Fixed-grid methods have been developed in order to address these challenges. Apart from these, mesh-free methods have also been proposed to tackle these obstacles. On the basis of domain discretization, mesh-free methods, and interface coupling, the numerical techniques have been classified into several categories.

In Sec. III, the myriad occurrences of FSI in biology (i.e., aquatic lifeforms, aerial habitats, and cardiovascular biomechanics) have been discussed. The behavior of soft elastic structures in plants (leaves, trees, etc.) when subjected to wind loads has been illustrated. The behavior of aquatic lifeforms during propulsion through fluidic environments has been discussed briefly. The manipulative FSI and adaptive interaction of the highly deformable life forms have been elaborated. The inherent presence of FSI in soft robots (i.e., bionic flaps, microfluidic devices, and soft robotic swimmers) and their fundamental role-play that affects their overall system behavior and system dynamics have been discussed. The physical interaction between stimuli-responsive soft elastica and the surrounding fluid in applications like microfluidics and lab-on-a-chip devices, as well as biomedical sectors such as targeted drug delivery and laparoscopy has been reported in light of FSI and their numerical modeling; examples ranging from artificial cilia to soft robotic swimmers (e.g., octobots) have been mentioned.

Nevertheless, in spite of these extraordinary achievements, the high-fidelity numerical modeling of three-dimensional fully coupled physics (during FSI) and cross-domain energy transduction (during multi-domain energy coupling) is yet to be thoroughly understood. Below, we, therefore, list a few possible areas of research for further exploration in this field:

- *Machine-learning models:* With an exponential development and a sharp progress in the computer resources and availability of enormous generated data, machine learning-based data-driven modeling techniques have immense possibility to significantly improve the prediction and understanding of FSI in coupled complex soft robotic systems.⁵⁹⁴ For example, a mathematical formulation for low Reynolds number hydrodynamics has been reported recently for helices and soft rotary propellers using a helical mode of swimming.⁵⁹⁵ The resistive force theory-inspired neural network-based hydrodynamic model has been proposed and trained using datasets from numerical simulations of the regularized Stokeslets method. The machine learning resistive force theory approach not only predicts the hydrodynamic force distribution but also accounts for the swimming kinematics of the slender structure. Although the accuracy remains similar (and comparable to the classical physics-based approach), the computational speed of the data-driven approach outnumbers the physics-based counterpart, thereby providing a computationally inexpensive framework for the estimation and comprehension of FSI and kinematic performance of soft robotic swimmers.
- *Hydroelastic FSI:* Problems involving large structural displacements, dynamically changing interfaces,⁵⁹⁶ and nonlinear free surface flows are difficult to model using conventional numerical methods.⁵⁹⁷ In an attempt to address these violent FSI, a robust numerical route has been developed recently where the flow field has been modeled using SPH, and the structural domain has been analyzed using a finite element method;⁵⁹⁸ the new approach has been termed as the smoothed particle element method.
- *Particle-based FSI models:* In order to overcome the current limitations faced by state-of-the-art FSI models, mesh-free methods such as particle-based formulations have been recently proposed for high-fidelity numerical simulations of heart valves and complex cardiovascular biomechanics.⁵⁹⁹ A monolithic approach has been adopted in this study (to evade the numerical problems faced by asynchronously coupled partitioned approaches), where the fluid and solid domains (represented as particles) have been solved simultaneously as a unified system using SPH. The adaptive, flexible and simple mathematical methodology employed in this approach makes it further suitable to be used for futuristic real-time modeling of heart-related disease⁶⁰⁰ and diagnosis.⁶⁰¹
- *Cellular mechanotransduction:* Cells respond to their dynamic environments by transduction of mechanical forces from the environments to the tissues and organs. These physical mechanisms and underlying phenomena of mechanotransduction are indeed fundamental for most biological processes, and yet they remain elusive till date.⁵³ In a recent study, FSI modeling of these phenomena has provided some light along this direction.⁶⁰² However, much remains to be decoded for a thorough understanding of the cellular biophysics and questions regarding the origin and processes of life.
- *Brain tissue/fluid modeling:* The dynamic interplay between the cerebrospinal fluid and the brain during traumatic injuries has been recently studied using FSI models.⁶⁰³ A thorough investigation on the influence of brain atrophy on brain injury during multiple sclerosis has been reported using FSI modeling for a higher accuracy.⁶⁰⁴ A mathematical formulation has also been recently developed for studying the dynamic FSI and intracranial

fluid dynamics between the pulsatile blood flow, brain, and cerebrospinal fluid.⁶⁰⁵

ACKNOWLEDGMENTS

The work is carried out within the research program of the Centre for Data Science and Systems Complexity (DSSC), Faculty of Science and Engineering, University of Groningen.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

R. Pramanik: Conceptualization (lead); Writing – original draft (lead).
R. W. C. P. Verstappen: Supervision (equal); Writing – review & editing (equal).
P. R. Onck: Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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